

EL AMAL website delivery and administration handbook

Delivery record and practical staff guide

Prepared for Nour Abulnasr and the EL AMAL team

Status date 29 September 2026

What is live, how it works, how to operate it and what remains.

Approximately 90 percent of the recorded scope is complete. The real catalogue is published. Customer email and several operational integrations still need activation.

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This handbook explains what has been built, what the client can use today, how staff operate it, and what remains before full operational launch. It serves two readers: the business owner who needs clear instructions and the developer who needs implementation details, reasons and evidence. The current website is a bilingual industrial instrument catalogue and quotation platform. It is approximately 90 percent complete against the recorded scope; customer email activation and some operational work remain.

Use the public website to review the actual catalogue. Use the private administration, inventory and reporting tools with your staff account. A published catalogue entry describes a model or family; an exact SKU identifies a supplied configuration; a stock receipt records counted physical units. These are three separate records. Publishing a product does not create warehouse stock or promise a reservation.

The handbook records the release serving the client review address when it was prepared. Application commit 7f7326704305012e9637702b3112b5c85f554833 was deployed and verified; documentation checkpoint f751ef5 records its evidence. Historical project notes contain earlier counts and disabled features. The current state described here takes precedence over those historical checkpoints.

How to use this handbook

- Business owner: start with Delivery status and Where to go, then read the public experience and the remaining work register.
- Catalogue editor: read Catalogue administration, including publication gates, source evidence and the distinction between model availability and SKU stock.
- Sales and warehouse staff: use the administration procedures, stock examples and troubleshooting instructions. Review the permissions table before assigning roles.
- Developer or successor: use the architecture, API, security, email, backup, deployment and verification chapters, then the source map and environment reference.

Each chapter explains the user benefit, the mechanism and its purpose. A statement that a feature is implemented does not mean its external service is activated. The remaining work register names who must supply information, who implements the work and how completion will be checked.

Delivery status

The most recent estimate is 88 percent weighted completion, rounded to about 90 percent with roughly five percentage points of uncertainty. It measures completed scope. It is not a security score, an SEO score, a design award prediction or a promise of a delivery date. A critical unfinished capability can matter more operationally than its numerical share.

Workstream	Weight	Estimated completion	Practical position
Foundation and hosting	15%	95%	Website, CMS, roles, separate databases and deployments work. Recovery delivery and staff onboarding remain.

Workstream	Weight	Estimated completion	Practical position
Interface and motion	20%	90%	Bilingual responsive interface and approved entrance are built. Final brand content and broader acceptance remain.
Catalogue and content	15%	85%	All supplied catalogue groups are published. Exact configurations and final technical acceptance remain.
Enquiries and email	20%	80%	Private workflow and queues are implemented. Real submission and mail activation require sender and scheduler setup.
Inventory	10%	95%	Stock controls and audited operations are implemented. Real SKUs, counts, policy and training remain.
SEO security and release	20%	85%	Technical SEO, protection, reporting and restore evidence exist. Field performance, offsite backups and wider acceptance remain.

The estimate above is the existing project estimate, not a new independent audit of every proposal line. The detailed remaining register also identifies original proposal items that have not become live features, including broader document uploads and visitor interaction analytics. No completion percentage should conceal those gaps.

Live and usable today

The stable English and Arabic website includes the real catalogue, model search, filters, product information, genuine manufacturer images, specifications and technical document links. Visitors can build a quote basket and prepare a model and range request. Staff can sign in to the hosted CMS, manage permitted records, open the private stock console and view demand reports. The stock tools are available, although no exact production SKU definitions or opening quantities have been supplied.

Eligible public pages are technically indexable. The current sitemap and crawl checks cover 342 public pages. A successful technical crawl establishes that the tested pages and metadata are reachable and internally consistent; it does not establish that Google has indexed them or that they rank for a target query.

Implemented but awaiting activation

Public enquiry submission, customer confirmation email, staff notification email and emailed staff password recovery remain disabled. The confirmed receiving inbox is mohamed.sorour8@icloud.com. A receiving address does not prove ownership of a sending domain. Sender verification, provider credentials and a sufficiently frequent delivery worker must be configured and tested before those switches are enabled.

Private customer photo handling is deployed and configured, but the normal customer must first complete email verification. That means the normal end to end customer photo journey is also waiting for mail activation. Internal development tests do not substitute for a real customer receipt and confirmation test.

Work still requiring implementation or operational setup

Larger PDF and Excel uploads with independent scanning are not implemented. Current support is limited to three private JPEG or PNG photos of up to 2 MiB each. Automatic offsite backup scheduling is not active. Wider screen reader and device acceptance, sustained performance measurements and final operational acceptance remain. External uptime monitoring has been deliberately deferred by Nour and should not be repeatedly requested.

Where to go

Task	Address or location
English website	https://el-amal-sigma.vercel.app/en
Arabic website	https://el-amal-sigma.vercel.app/ar
Product catalogue	https://el-amal-sigma.vercel.app/en/products
Quote basket	https://el-amal-sigma.vercel.app/en/quote
Direct model request	https://el-amal-sigma.vercel.app/en/rfq
Staff sign in and CMS	https://el-amal-sigma.vercel.app/admin
Stock operations	https://el-amal-sigma.vercel.app/staff/inventory
Sales demand reports	https://el-amal-sigma.vercel.app/staff/reports
Source repository	https://github.com/nourabulnasr/EL-AMAL
Working branch	codex/el-amal-foundation
Local project	C:\Users\noura\OneDrive\Documents\ChatGPT\EL-AMAL 4

Replace /en with /ar for the corresponding Arabic public journey. The administration and staff tools share the existing staff account. Localhost on port 3004 is the developer's test server; the client and staff use

the public HTTPS addresses above. Keep the stable client URL bookmarked instead of an individual deployment URL.

What the project includes

EL AMAL helps engineers identify measurement instruments and helps procurement teams describe what they need. The site is designed around product identity, technical information and quotation requests. It does not include public prices, checkout, payment processing, customer accounts, ERP synchronization, automatic purchasing or AI engineering advice. Those were outside the initial scope. No such capability should be inferred from a basket icon or a product page.

The initial brief proposed six implementation increments: foundation, catalogue slice, RFQ slice, stock slice, completion and release candidate. Later progress reports use six workstreams instead. These are two ways to organize the same project, not twelve phases. We are in completion and release acceptance across those workstreams. Core construction is substantially complete, while several operational dependencies remain.

The original proposal mentioned four industry pages. The later client requirements prioritized oil and gas and general industry; those two are the implemented industry routes. Additional pharmaceutical, food and beverage, and construction or EPC content needs to be reconciled with the final agreed scope rather than described as already delivered. The proposal also mentioned a factory location; a verified address, hours and location presentation remain dependent on business information.

How the project developed

Milestone	What changed	Result
Foundation and first review link	Next.js and Payload established with Neon and Vercel	Client could review the same hosted site throughout development.
Client requirement implementation	Type and application filters, model and range RFQ, two industry pages and guarded business contact components	The structure follows the client's quotation workflow.
Navy design and interaction work	Requested navy colors, white type, orange details, focus and hover treatments, instrument motion	One consistent public visual system with mobile and motion safeguards.
Branded entrance	Instrument inspired loading presentation, dismissal and replay	Approved entrance without blocking page reading indefinitely.
Real catalogue publication on 28 September	All 29 supplied main pages imported as 151 model groups	Actual images, specifications and documents replaced review fixtures.
Operations and technical SEO on 29 September	Stock service, encrypted photos, delivery maintenance, recovery safeguards and full public crawl	Private workflows and indexable real content became available.

Milestone	What changed	Result
Completion release on 29 September	Partial stock operations, blocked balances, count reviews, reports, encrypted restore rehearsal, font and loader changes	Broader staff control, recovery evidence and smaller public payloads.

The source repository contains the implementation and migration history. PROGRESS.md contains dated release evidence. Database records live in Neon and change independently of Git. A pushed commit preserves code; it does not copy the current enquiries, stock ledger, users or unpublished CMS changes.

The four parts of the design

The focal subject is an industrial instrument and a clear route to model selection. Light and contrast come from white type and restrained orange details against the requested #010736 and #091540 surfaces. Framing uses editorial headings, generous space and an off centre instrument composition. The intended feeling is engineering confidence and clarity. These choices carry through the catalogue and staff presentation, while technical tables remain readable.

The requested animation list was treated as a set of possible techniques. The implementation uses lightweight CSS interactions and Motion where it benefits the hero; it does not install every animation library mentioned. The approved loading entrance is decorative. Real route loading still follows the application's actual work. Performance, reduced motion and touch usability limit how much motion each screen needs.

EL AMAL handbook public website catalogue design and search

Evidence date: 29 September 2026. This chapter describes application commit 7f7326704305012e9637702b3112b5c85f554833, deployed as dp1_4iYTN8joqJgEfHa7L2GZzR4pn2bA. The documentation checkpoint preceding this handbook is f751ef5. The stable public address is [EL AMAL English](#) and [EL AMAL Arabic](#).

This is documentation of the existing implementation. No application code, deployment, database or external account was changed to produce this chapter. Statements about live checks refer to the saved release evidence, not a fresh live audit. Repository-relative source paths below resolve from C:\Users\noura\OneDrive\Documents\ChatGPT\EL-AMAL 4.

1 What the public website does

EL AMAL helps an engineer identify an industrial instrument and helps a purchasing team assemble the details needed for a quotation. Visitors can browse a real bilingual catalogue, search a model reference, narrow the results, open manufacturer documents, collect models and quantities, and prepare an enquiry. They can also start a direct request for a model that is not listed.

The site is a quotation catalogue. There are no public prices, checkout, payment processor, customer accounts, automatic purchasing, ERP integration or AI engineering recommendations. These were

outside the initial proposal. Adding an item to the basket does not reserve stock or place an order. A catalogue model can represent a family with many possible configurations; it is not automatically an exact orderable SKU.

The real catalogue and eligible search pages are publicly available. Visitor enquiry submission, real customer email delivery and password-recovery delivery remain disabled until the sender and frequent delivery scheduler are ready. A visitor can prepare and review a request but must not be told that a review screen has sent it. Private staff operations are implemented separately from these public readiness switches.

The current public header consequently displays: “Client review - enquiries and stock reservations are not yet available,” with an Arabic equivalent. This notice describes visitor-facing availability; it does not mean the private stock ledger or staff inventory console is absent. Search indexing and email activation are independent controls.

Source: README.md; PROGRESS.md current milestone; src/components/header.tsx;
src/components/enquiry-preview.tsx; src/components/direct-rfq.tsx; src/lib/customer-readiness.ts;
docs/kickoff-2026-09-17.md.

2 Public pages and navigation

Complete route families

Every content route below exists in English and Arabic. Substitute en or ar for {locale}. Product details are generated from published CMS records, not 151 separately authored page files.

Route	What the visitor sees	Search treatment
/	Redirect to /en	Entry redirect, not a separate sitemap page
/ {locale}	Homepage, model search, category entry points, selected instruments, quotation process and industry links	Eligible when production indexing is enabled
/ {locale} /products	All 151 published model groups, filters and first 24 results	Eligible
/ {locale} /products?page=2 through ?page=7	Remaining catalogue pages; page 7 has seven entries	Each valid page has its own canonical and sitemap entry
/ {locale} /categories/pressure and ?page=2	47 pressure entries	Both pages eligible
/ {locale} /categories/temperature and ?page=2, ?page=3, ?page=4	87 temperature entries	All four pages eligible
/ {locale} /categories/accessories	17 valves/accessories entries	Eligible
/ {locale} /products/cms-N	One published model-group detail, real image, availability date, technical rows and document links	All 151 existing IDs have eligible EN/AR pages; exact list in Appendix A

Route	What the visitor sees	Search treatment
<code>/{{locale}}/industries/oil-gas</code>	Oil and gas application briefing/checklist, direct RFQ and related application filter	Eligible
<code>/{{locale}}/industries/general-industry</code>	Factory/utility/replacement briefing/checklist, direct RFQ and related filter	Eligible
<code>/{{locale}}/about</code>	Current approach and quotation process; conditional WIKA relationship evidence	Eligible
<code>/{{locale}}/contact</code>	How to begin a request; current submission status; conditional phone/WhatsApp actions	Eligible
<code>/{{locale}}/resources</code>	Technical enquiry preparation checklist	Eligible
<code>/{{locale}}/quote</code>	Browser-saved basket, quantities, removal and enquiry preparation	<code>noindex, nofollow</code> ; omitted from sitemap
<code>/{{locale}}/rfq</code>	Direct model/quantity/range request and review	<code>noindex, nofollow</code> ; omitted from sitemap
<code>/{{locale}}/verify</code>	Token-based enquiry confirmation; optional photos after real confirmation	<code>noindex, nofollow, no-referrer</code> ; omitted from sitemap
<code>/{{locale}}/products?q=...&category=...&type=...&application=...</code>	Any supported combination of model search and filters	<code>noindex</code> ; production allows following links
<code>/sitemap.xml</code>	Generated bilingual discovery list	342 URLs in the final audit
<code>/robots.txt</code>	Crawler rules and, when eligible, sitemap location	Public machine-readable endpoint
<code>/social-image</code>	Stable 1200 × 630 PNG sharing artwork	Public asset
<code>/favicon.ico, /icon.svg</code>	Existing navy/orange/white A monogram	Public assets
<code>/images/products/*.png</code>	151 original manufacturer product images	Public assets

The locale `opengraph-image.tsx` file also participates in Next's generated image convention. Nested-page metadata deliberately points to `/social-image`, whose path is stable rather than relying on Next's generated suffix. These are image routes, not separate editorial pages.

The 342 sitemap pages reconcile exactly: for each language there are six home/information/industry pages, seven catalogue pages, seven category pages and 151 product pages: $2 \times (6 + 7 + 7 + 151) =$

342. The count does not include quote/RFQ/verification pages, filter combinations, assets or administrative routes.

Unknown product/category/industry/information identifiers call `notFound()`. Unknown locales are rejected. There is a localized not-found view inside the site and a standalone bilingual global 404 fallback. Because Next may begin streaming before a missing record is discovered, two missing nested routes in the saved SEO audit returned HTTP 200 with explicit `noindex`; `/xx` returned HTTP 404. Do not describe every missing nested route as returning 404.

There are no implemented public `/privacy`, `/terms`, `/delivery`, `/services`, `/industries/pharmaceutical`, `/industries/food-beverage` OR `/industries/construction-epc` pages. The information route allows only about, contact and resources.

Source: `src/app/(redirect)/page.tsx`; `src/app/(site)/[locale]/`; `src/content/information.ts`;
`src/content/industries.ts`; `src/lib/seo-discovery.ts`; `src/app/social-image/route.tsx`;
`src/app/global-not-found.tsx`; `artifacts/2026-09-29/completion/seo-final.json`.

Header footer and contact bar

The header links the EL AMAL wordmark to the current language homepage. Main navigation contains Catalogue, Industries, Request a quote and About. Industries opens the homepage industry section; its two entries then open the dedicated industry pages. A language switch and quote-basket link sit alongside the navigation. The basket badge counts distinct model lines, not total requested units.

The header is sticky. An `IntersectionObserver` changes its scrolled appearance without a continuous custom scroll listener. Below 1024px the navigation becomes a Menu disclosure; at the smallest widths the language and basket actions form their own row. The toggle exposes `aria-expanded` and `aria-controls`; choosing a menu link closes it, and Escape closes it and returns focus to the toggle. A no-JavaScript style keeps navigation links visible and hides the unusable toggle.

The footer repeats catalogue, basket, About, Resources and Contact links, includes a localized introduction replay button, and shows the current year. A separate contact bar always offers the RFQ route. Telephone and WhatsApp links appear only when valid approved numbers are configured; they are not placeholders.

`BUSINESS_PHONE` and `BUSINESS_WHATSAPP` accept international numbers beginning with + and a nonzero country code, followed by 7-14 more digits. The phone becomes a `tel:` link. WhatsApp removes the plus sign for its `wa.me` URL. The direct RFQ can open a prepared WhatsApp message only when that number is present. Opening the draft is not sending a message.

Source: `src/components/header.tsx`; `src/app/(site)/[locale]/layout.tsx`;
`src/components/business-contact.tsx`; `src/lib/business-contact.ts`; `src/components/direct-rfq.tsx`.

What changed from the original proposal

The 17 September kickoff proposed four industry pages: oil and gas, pharmaceutical, food and beverage, and construction/EPC. The 24 September client-requirements record explicitly says the client screenshot takes priority over the previous phase sequence, and specifies the two implemented industries: oil and gas and general industry. The content file, product application choices and sitemap all use those two. This handbook records the later implementation scope; it does not count four completed industry pages or silently assume the three other original sectors exist.

The original “up to 250 models / 500 supplied SKUs” was a content allowance, not evidence that 250 real models or 500 stock items were supplied. The later authorized publication covers all 151 model groups on the 29 actual photographed main pages. There is no missing batch implied by subtracting 151 from the original allowance. Exact stocked configurations and opening balances remain separate client inputs.

Source: docs/kickoff-2026-09-17.md; docs/client-requirements-2026-09-24.md;
docs/catalogue-publication-2026-09-28.md; src/content/product-options.ts.

3 How the real catalogue was built

Sources counts and what they mean

Evidence	Verified recorded quantity	Meaning
Original photographed main pages	29	17 in-stock and 12 out-of-stock source photos; partial neighbouring pages excluded
Published entries	151	Printed cards/model groups, including grouped families; not 151 exact SKUs
In-stock / out-of-stock labels	90 / 61	User-reported main-page classification, dated 27 September 2026
Measurement categories	47 pressure / 87 temperature / 17 accessories	Totals reconcile to 151
Original manufacturer images	151	One image path for every published group; 151 actual PNG files
Bilingual specification rows	560	Each row has English/Arabic labels and values; not 560 separate rows per language
Translated terms checked	486	Publication report's terminology/numeric-fidelity check
Datasheet identities/downloads	159	158 verified codes plus one explicitly recorded official-link/code mismatch
Entries with official datasheet links	150	The remaining M12 cable entry links the manufacturer catalogue at page 29
Document references in publication JSON	164	Families can have several documents and documents can recur; 160 distinct URLs including the catalogue fallback
Application tags	33 oil/gas / 90 general industry	Manufacturer-evidenced classification; tags can overlap and are not suitability guarantees

The counts were read from the publication report and checked against publication.json: 151 products, 151 unique image paths, 560 technical rows, and the 90/61 split. datasheet-links.json records 159 codes, 158 normally verified codes, one official-link/code mismatch, zero unresolved codes, 149 cards with

complete conventional datasheets and 150 with official datasheet links. The distinction explains why “150 entries have links” is not identical to “150 entries have perfectly matching printed datasheet codes.”

Nour's explicit rule is that every model on a main photographed page inherits that folder's stock label. Pen marks do not select stock. This rule supersedes earlier intake uncertainty. It must not be re-asked or silently reinterpreted. The label is a dated family-level report, not a live count, factory commitment, configuration guarantee or reservation.

The images came from the matching official WIKA portfolio, not AI replacement product photography. They show the model family; a supplied variant can differ. The homepage's decorative instrument remains an illustration and is explicitly captioned as such. It must not be confused with the actual product photographs.

Source: docs/catalogue-publication-2026-09-28.md; catalogue/2026-09-27/publication.json; catalogue/2026-09-27/datasheet-links.json; catalogue/2026-09-27/source-manifest.json; public/images/products/; src/components/hero-instrument.tsx.

Extraction translation and publication path

The source workflow retains photographed originals privately, records filenames/dimensions/hashes, indexes each main page, extracts matching official catalogue material, translates the selected technical information, verifies manufacturer document identities, and builds importable publication records. The reproducible files include `extract-official.py`, `official-extraction.json`, `prepare-content.mjs`, `content-source.json`, `translation-input.json`, `translations.json`, `verify-datasheets.mjs`, `datasheet-links.json`, `build-publication.mjs` and `publication.json` under `catalogue/2026-09-27/`.

The importer validates before writing, uses stable external IDs, and records a backup of affected records. Dry run is its default; writes require the explicit chosen target. The recorded development import resumed with 64 unchanged and 87 new records, without duplicates. Hosted publication created and verified all 151 records. The 28 September live check compared SHA-256 hashes for all 151 served PNG files against local originals.

A product can appear in the public projection only when it is published and has both-language names and descriptions, a source reference, reviewer identity, parseable review date and reuse-rights confirmation. The public object is rebuilt from an allowlist. Internal source/reviewer records, draft versions and exact SKUs do not pass through to the public page. Categories without projected products are omitted. CMS mode never silently falls back to plausible demo products when reads fail or return no products.

This is a publication-control mechanism, not a manufacturer's certification of every specification or a substitute for the client's final technical/Arabic review. Original photos/PDFs and operational artifacts are ignored by Git; a clone is not a backup of those private source files. Approved extracted product PNGs are public assets.

Known source corrections were documented rather than hidden: TFT35's printed TE76.18 resolves to TE67.18; BA's PV32.22 resolves to PV32.21; the official PGT21 document has inconsistent printed PV21.02 headers, so its public title identifies WIKA PGT21; the missing A-1200 negative sign was restored from PE81.90; an ambiguous generic HPNV low-pressure option was omitted in favour of supported nominal information. None of these corrections creates an offered configuration or certification claim.

Source: scripts/import-catalogue.ts; src/lib/import.ts; src/lib/access.ts; src/lib/public-catalogue.ts; src/lib/catalogue-details.ts; docs/catalogue-publication-2026-09-28.md; catalogue/2026-09-27/README.md.

Search filters and pagination

Search works across the model reference, both-language name and both-language description. It is not a separate search service. Normalization applies Unicode NFKC, lowercase, removes punctuation/spaces and Arabic vowel marks/tatweel, and normalizes the common alef variants. An exact normalized model match sorts first. Input is capped at 120 characters. This lets model references survive punctuation and mixed-script differences without pretending to offer semantic engineering advice.

Four inputs combine: text query, measurement category, instrument type, and application. Types are pressure gauges, pressure transmitters, pressure switches, temperature instruments, and valves/accessories. Applications are oil/gas and general industry. Filters submit through ordinary URL forms, so results can be bookmarked and remain readable without JavaScript. Active filter labels and reset links make the current state visible. A search/category chip can remove that specific condition; type/application labels are shown and can be changed through their selects or the full reset.

Pagination renders 24 cards per page with ordinary Previous/Next anchors. Filters are preserved in navigation. Invalid, noninteger, unsafe or out-of-range page values resolve to page 1. Category-only navigation can use a dedicated `/categories/<key>` path; combinations with query/type/application use `/products` query parameters. Canonical generation normalizes valid page state and excludes page 1's redundant `page=1` parameter.

The zero-result view suggests shortening the model or resetting filters. An empty real catalogue instead says it is being prepared; it does not substitute synthetic content. A direct-RFQ link covers a model outside the catalogue. Card descriptions are visually limited to two lines; full descriptions and the technical table remain available on the product detail.

Source: `src/lib/catalogue.ts`; `src/components/catalogue-view.tsx`; `src/components/product-card.tsx`; `src/content/product-options.ts`; `src/lib/seo-discovery.ts`; `src/app/(site)/[locale]/products/page.tsx`; `src/app/(site)/[locale]/categories/[slug]/page.tsx`.

Product detail behavior

Each detail opens with a breadcrumb, manufacturer image, category link, isolated model code, localized title/description, dated availability and Add to quote. The page explicitly says that configuration and current availability need confirmation during quotation. The technical section explains that ranges/options describe the family, then renders manufacturer-document buttons and a definition list containing the model, category, source status and bilingual specification rows.

Images use `next/image`, known dimensions, responsive sizes, and `object-fit: contain` so the whole instrument remains visible. Card images are lazy; the main detail image is eager with high fetch priority because it is prominent initial content. Fixed photo stages and declared dimensions reduce image-driven layout shifts. Document links are HTTPS and open with `noopener noreferrer`; missing documents have an explicit message. Product availability is a textual status plus a localized date rather than color alone.

No price, stock quantity, product rating, certification badge or authorization claim is inferred. Final suitability, revision, variant, availability and commercial terms need human confirmation. There is no gallery of invented alternative views or 3D product viewer.

Source: `src/app/(site)/[locale]/products/[slug]/page.tsx`; `src/components/product-image.tsx`; `src/components/product-availability.tsx`; `src/lib/catalogue-details.ts`; `src/lib/public-catalogue.ts`.

4 Basket direct RFQ and confirmation experience

Basket sequence

1. Open a product and select **Add to quote**. Each click adds one unit to that model's existing line or creates a new line. A status message confirms the interface action.
2. Open the header's quote basket. Change quantities from 1 to 9,999 whole units, remove a line, or return to the catalogue.
3. Add name, work email and company, with optional application/configuration notes. Review the details before any submission action.
4. In the current gated production state, the interface truthfully remains a form preview. The separate authorized staff test-save function is available only in demo catalogue mode, not in the current real CMS catalogue mode. Visitor review does not save a real enquiry or send email.
5. When prerequisites are configured and the independent intake switch is deliberately enabled, the customer submission control can save the immutable request and queue confirmation. Confirmation must happen before the downstream customer workflow; adding a basket item never reserves stock.

Only product IDs and quantities are saved in browser `localStorage`. Demo and CMS baskets use different keys: `e1-amal-preview-basket-v1` and `e1-amal-cms-basket-v1`. The provider removes records that no longer exist in the current projected catalogue, rejects malformed saved structures, limits baskets to 100 distinct lines, and synchronizes other tabs through the storage event. An unavailable storage mechanism produces a warning and leaves the current session usable. The browser catalogue payload contains only each product's ID, model and bilingual name; all technical detail remains server-rendered.

The basket is device/browser storage, not a logged-in cross-device cart. Contact details are component state, not persisted basket data; reloading can lose an unfinished form. Editing/saving the basket requires JavaScript and the no-script message says so. Catalogue reading and product document access do not require the basket to work. A 100-line add limit exists in the library; the product button does not provide a separate visible full-basket message, so that extreme case is not claimed to have complete UX acceptance.

The basket enquiry uses inline localized validation, required labels, email direction isolation, explicit invalid-field descriptions and focus on the first invalid field. It moves focus to the review heading after successful validation, and back to the form when editing. Name, email, company and notes are bounded at 120, 254, 160 and 2,000 characters respectively.

Source: `src/components/add-to-quote.tsx`; `src/components/basket-provider.tsx`; `src/lib/basket.ts`; `src/components/quote-basket.tsx`; `src/components/enquiry-preview.tsx`; `src/lib/enquiry-preview.ts`; `src/app/(site)/[locale]/layout.tsx`.

Direct request sequence

The direct RFQ requires model, integer quantity, measurement range/unit, name, email and company, with optional notes. Model/range can describe an unlisted instrument. The form uses native required/email/number/pattern validation and limits model to 120 characters and range to 160. Review disables the input fieldset, shows that technical suitability/availability need confirmation, and offers Edit details. The optional WhatsApp action assembles the details into a URL-encoded draft for the visitor to send.

The activated customer control uses a request fingerprint plus an idempotency key, a busy lock, bounded request timeout and localized rate-limit/unavailable feedback. Retrying the same details is intended to avoid duplicate requests. A saved response displays a reference and says the confirmation email is queued, not delivered. Resend uses a returned receipt, a visible 60-second cooldown and eligibility wording. The server revalidates these limits and identities; client validation alone is not a trust boundary.

Source: `src/components/direct-rfq.tsx`; `src/components/customer-enquiry-submit.tsx`;
`src/components/test-enquiry-submit.tsx`; `src/lib/customer-service.ts`; `docs/customer-intake.md`.

Confirmation and attachments

The verification page reads the token from a URL fragment, shows an explicit Confirm action, and removes the fragment from browser history before the confirmation request. Statuses distinguish missing/expired/used links, rate limiting, uncertain failures, real customer verification and staff test confirmation. A test confirmation explicitly does not prove customer email ownership. Confirmation itself does not place an order or reserve stock.

A genuine customer confirmation may return a short-lived attachment grant and display the photo component. The implemented subset is up to three JPEG/PNG photos, 2 MiB each, with 8-megapixel processing limits, metadata removal, private encrypted handling, 30-day access and bounded logical storage. It is not the proposal's broader three-file PDF/JPEG/PNG/XLSX, 10 MB-per-file feature. The pre-submission "Drawings & technical documents" text is a placeholder for future broader support; do not describe it as an enabled PDF/Excel uploader. Real customer entry to this path remains constrained by the disabled public intake/email chain.

Source: `src/app/{site}/{locale}/verify/page.tsx`; `src/components/verify-enquiry.tsx`;
`src/components/enquiry-photo-upload.tsx`; `docs/private-enquiry-photos.md`; `docs/phase-status.md`.

5 English Arabic and mobile behavior

The locale layout sets `real html lang="en"/lang="ar" and dir="ltr"/dir="rtl"`; translation is stored in code/content fields rather than supplied by a runtime translation service. The same record identity links both languages. English uses Manrope body text and Newsreader display headings. Arabic uses Noto Sans Arabic for body/headings with its own line-height and size rules. Arabic titles are not forced into the English spacing metrics.

Model references, signed ranges, units and email addresses retain left-to-right isolation where needed. `BidiText` separates Latin/numeric runs inside Arabic prose with `<bdi dir="ltr">`; explicit model fields also use isolation. This matters for references such as grouped models, a minus sign in a temperature range, or connection dimensions. Technical review still needs to validate terminology; layout isolation is not translation certification.

With JavaScript, the language switch replaces the first locale segment, then preserves the current query string and fragment through a full document navigation. Thus an Arabic filtered catalogue can retain its filters when switching to English, and the same local basket survives the switch. The anchor itself preserves the path without JavaScript, but query/hash preservation is implemented in its click handler; do not claim the complete query-preserving switch in no-script mode. Unsaved form component state is not transferred between languages.

The layout starts as a vertical phone layout. Catalogue controls precede results; the filter becomes a sticky sidebar on a large screen. Product details stack and later become two columns. The final desktop catalogue uses two product columns beside the filter, while the homepage's selected-instrument

composition deliberately gives the first product a larger stage. Category tabs can scroll horizontally, product/model/table text can wrap, and on widths below 420px technical definition rows become one column. Wider layouts constrain the overall readable width instead of stretching indefinitely.

The important thresholds are 640px for several two-column arrangements, 768px for desktop-motion eligibility, 900px for editorial information-page composition, 1024px for full navigation/hero/filter layout, and 1500px for additional outer margins. Different components need different thresholds; there is no claim that all responsive behavior is controlled by one breakpoint.

Recorded browser checks include 320px grouped product details, 375/390px mobile views and 1440px desktop samples, both locales, language switching, filters, basket persistence and no horizontal page overflow in the sampled routes. These are useful samples, not coverage of every phone, browser, zoom factor or all 342 pages in a rendered browser.

Source: `src/app/(site)/[locale]/layout.tsx`; `src/app/(site)/[locale]/styles.css`;
`src/components/header.tsx`; `src/components/bidi-text.tsx`; `src/content/copy.ts`;
`artifacts/2026-09-29/launch/browser-audit.json`; `artifacts/2026-09-29/completion/final-browser.log`;
`docs/catalogue-publication-2026-09-28.md`.

6 Design and motion what is actually implemented

Visual direction

The selected direction began as **Precision in steel**, then adopted Nour's exact navy request on 27 September. Its four design layers are: the instrument and quotation task as the subject; bright white type and orange details against deep navy as the light/contrast; oversized editorial headings and an off-centre instrument as the framing; precision and confidence as the intended feeling.

The active background values are #010736 and #091540, white written content, and the existing orange accent `oklch(72% .14 55)`. Source CSS retains older steel/neutral rules before the later navy overrides; reading only the first `:root` block gives the wrong current palette. It is a mixed implementation: the final navy overrides use exact hex values, with OKLCH retained for the accent and some effects. It is not a uniformly converted OKLCH-only design system.

The final override deliberately makes body/muted text white, distinguishes adjacent navy surfaces with borders/spacing, and keeps dark glyphs on orange controls. Actual manufacturer images sit on white contained stages to preserve the complete photographed instrument. Metal shading inside the illustrative gauge is retained. The current A monogram/wordmark and supplied-content presentation are provisional identity work pending the final logo and approved company assets.

Manrope handles functional text; regular-weight Newsreader provides English display hierarchy; Arabic has Noto Sans Arabic. Hero headings use responsive sizing up to approximately 142px in the large English composition, while technical copy and labels are materially smaller. Most pages use generous section spacing, clear rules and editorial asymmetry rather than repeated equal-weight rounded cards. This describes implementation and intention; only Nour grants visual acceptance.

Source: `docs/quality-roadmap-2026-09-27.md`; `src/app/(site)/[locale]/styles.css`, especially the "Client palette - 27 September 2026" block; `src/app/(site)/[locale]/layout.tsx`; `src/components/instrument.tsx`; `PROGRESS.md` 20 and 27 September entries.

Hero depth and motion

The hero instrument is a layered HTML/CSS illustration: gradients, a case, dial, needle, stem and shadow. Perspective/rotation make it appear dimensional. There is no polygon mesh, WebGL scene or downloadable 3D model. Its caption expressly says “Illustrative study, not product photography.”

On a fine hover-capable pointer, at least 768px wide, and with normal motion preference, a dynamic import loads the interactive component. `motion/react` uses `LazyMotion` and separately loaded features. Pointer position drives spring-smoothed X/Y rotation; scrolling through the instrument region compresses its scale from 1 to 0.88 and moves it by up to 45px. The pointer spring uses stiffness 95, damping 22, mass 0.7. Leaving the instrument returns it toward neutral. A visible Pause motion / Enable motion control swaps the interactive component for the static instrument.

Mobile, coarse-pointer and reduced-motion visitors receive the server-rendered still instrument without loading this desktop Motion component. JavaScript-disabled visitors also retain readable static content. This is a bounded interaction around the instrument, not scroll interception for the whole site.

Source: `src/components/hero-instrument.tsx`; `src/components/interactive-instrument.tsx`;
`src/components/motion-features.ts`; `src/components/instrument.tsx`.

Introduction loading hover and focus

Feature	Implemented behavior	Limit or distinction
Branded introduction	Navy split panels, white editorial EL AMAL wordmark, orange gauge sweep, localized caption/Skip, footer Replay	Decorative entrance; no fake network percentage
Entrance duration	Desktop CSS sequence 1.8s; mobile below 768px 1.15s; JS fallback dismissal 2.2s	Runs on document mount/replay, not every client navigation; no once-per-session storage flag
Dismissal	Pointer press, any keyboard input, focus entry, wheel/touch movement and explicit Skip can dismiss	Does not lock body scroll or take focus for decoration
No-JavaScript / reduced motion	Introduction/replay hidden; global animation/transition reductions; route needle static	No mandatory animated gateway to content
Actual route loading	Localized <code>role="status"</code> , polite live region and <code>aria-busy</code> , with gauge needle	Fallback reflects Next route work; not a simulated duration or progress estimate
Header	Sticky, darker/translucent scrolled state with CSS backdrop blur	CSS glass appearance, not a liquid-glass library
Buttons	Fine-pointer lift, one-pass sheen, arrow displacement; immediate press response	CSS spring-shaped <code>linear()</code> easing, not physics simulation for every element
Links/categories/cards	Navigation underline, category text/arrow movement, image-stage lift, active-filter feedback	Hover movement is opt-in for fine pointers; touch retains press feedback

Feature	Implemented behavior	Limit or distinction
Form focus	Visible orange outline/boundaries and error treatment	Keyboard focus is explicit; full assistive-technology acceptance remains open

The same-segment `loading.tsx` does not unblock data already awaited in the locale layout. The current layout awaits the fresh catalogue before returning its full shell. That is an identified performance tradeoff. A whole-site Suspense experiment was not applied because its tested hidden late segment required JavaScript to reveal real content; keeping the no-script catalogue readable took precedence in the existing implementation.

`package.json` includes Motion 12.43.0. It does not include Vanta, Three.js, React Three Fiber, GSAP, Lenis, Anime.js, a liquid-glass package or a logo-morph package. There is no video hero, procedural particle scene, original 3D modelling, scroll hijacking, true liquid logo or 3D product viewer. Requested visual outcomes must be described in terms of the actual CSS/Motion implementation, not attributed to unused libraries. The split-panel introduction is an EL AMAL composition informed by a reference entrance; it is not evidence that the reference site's implementation was copied.

Source: `src/components/site-intro.tsx`; `src/app/(site)/[locale]/loading.tsx`;
`src/app/(site)/[locale]/styles.css`; `src/components/header.tsx`; `package.json`;
`docs/performance-completion.md`; PROGRESS.md 25/27 September interaction entries.

7 Technical SEO and discoverability

Twelve implemented approaches and why

These twelve groups describe concrete implementation choices, not a claim that there are twelve ranking factors or that the site has earned a particular ranking.

1. **Server-render the real catalogue.** Product names, descriptions and technical content reach readers/crawlers as HTML, with no-script readability preserved. This reduces dependence on browser JavaScript for discovering the business's actual content.
2. **Build a crawlable hierarchy.** Home links to categories and selected products; catalogue/category pages link to detail pages; breadcrumbs support return navigation. Discovery does not depend only on search boxes.
3. **Give pagination real URLs.** Native Previous/Next anchors and a separate canonical for each valid page expose all 151 records, including later catalogue/category pages. Pagination is not collapsed to the first page.
4. **Exclude faceted search duplicates.** Search/category/type/application query results use `noindex` while their product links remain followable in eligible production. Dedicated category routes retain useful indexable landing pages.
5. **Generate page-specific metadata.** Localized titles and descriptions describe the actual page/model and page number. Canonicals identify its normalized URL rather than relying on a generic site title.
6. **Connect language equivalents.** English/Arabic reciprocal alternates, English x-default, correct document language/direction and distinct canonicals clarify the bilingual relationship without treating Arabic as duplicate English content.

7. **Describe facts with structured data.** Website/organization, page/breadcrumb, collection/list and real CMS Product entities describe what is actually present. Fictional prices, reviews and authorization claims are excluded.
8. **Gate search exposure by environment.** Demo/development/previews remain noindex; private/admin/API and quotation utility routes have permanent exclusions. Production indexing is a separate explicit decision from activating customer mail.
9. **Generate the sitemap from eligible records.** Both languages, valid pagination, nonempty categories and published products are included, with real image URLs and no invented last-modified dates. Drafts/internal data are excluded.
10. **Provide stable sharing and brand assets.** Explicit Open Graph/Twitter images and repaired favicon/icon routes prevent missing nested-page previews and broken brand assets. These are presentation/discovery basics, not ranking guarantees.
11. **Preserve content provenance and truthful claims.** Reviewed bilingual publication fields, actual manufacturer photos/documents and conditional WIKA relationship evidence limit unsupported commercial/technical assertions. Technical accuracy still needs final client acceptance.
12. **Audit the entire intended discovery set.** The bounded read-only crawler verifies route coverage, metadata, alternates, schema and assets, and reports partial-crawl failure rather than treating a sample as a full audit. Google indexing/traffic and field performance remain separate observations.

The subsections below give exact behavior and source references for these approaches.

Metadata contract

All indexable public page families use `generateMetadata` with the shared `pageMetadata` helper. Home titles are absolute, for example `EL AMAL | Industrial instrumentation`; other pages use the layout's `%s | EL AMAL` template. A product title combines its model and localized name. Descriptions come from the relevant bilingual page or product content. Valid page numbers are reflected in paginated titles.

Each helper-generated page has its own absolute canonical, English and Arabic language alternatives, and English `x-default`. Alternatives preserve the normalized page path/query. A canonical identifies the preferred URL for that page; it is not a promise of indexing. Product IDs are shared across languages. Valid pagination uses a self-canonical; it is not all collapsed to page 1.

Open Graph sets type `website`, EL AMAL site name, localized title/description, absolute page URL, `en_GB` or `ar_EG`, the reciprocal alternate locale, and the `1200 × 630 /social-image`. Twitter uses `summary_large_image` with title/description/image. The shared artwork contains English EL AMAL campaign wording; Arabic metadata does not imply a separately translated Arabic image. `GOOGLE_SITE_VERIFICATION`, if supplied, emits Google's verification meta token. No Search Console ownership or submission is implied by merely supporting that variable.

The helper's coverage should not be overstated: `/verify` deliberately supplies its own noindex/referrer metadata rather than the full public-page metadata contract, and standalone error pages are separate. Noindex utility pages are not intended to be SEO landing pages.

Source: `src/lib/page-metadata.ts`; `src/app/(site)/[locale]/layout.tsx`; the individual `page.tsx` files; `src/app/(site)/[locale]/verify/page.tsx`; `src/app/(site)/[locale]/opengraph-image.tsx`; `src/app/social-image/route.tsx`.

Production gate and exclusions

Indexing is allowed only when all four conditions are true: `SITE_INDEXING_ENABLED=true`, `CATALOGUE_SOURCE=cms`, `CMS_ENABLED=true`, and `VERCEL_ENV=production`. The canonical origin comes from a valid credential-free HTTPS `SITE_URL`; the current fallback/stable origin is `https://el-amal-sigma.vercel.app`. A preview does not become indexable merely because the switch was copied into its environment.

When the production gate is false, all routes receive `X-Robots-Tag: noindex, nofollow`, the layout metadata is `noindex/nofollow`, and the sitemap is empty. In eligible production, private administration/staff/API routes and EN/AR quote/RFQ/verification routes retain permanent `noindex, nofollow` headers. A `/products` request with any `q`, `category`, `type` or `application` key receives `noindex, follow`; even an explicitly present empty filter parameter is excluded. Plain pagination alone is indexable. The public metadata mirrors these distinctions.

`robots.txt` allows `/`, disallows `/admin`, `/staff` and `/api/`, and lists the sitemap only when the production gate is enabled. It does not disallow quote/RFQ/filter pages, because their `noindex` instructions need to be crawlable. Robots exclusions are discovery policy; access control and private-response protections are separate server mechanisms. Environment changes affecting configured headers/layout metadata require redeployment.

Source: `src/lib/site-policy.mjs`; `next.config.mjs`; `src/app/robots.ts`; `src/app/sitemap.ts`; `src/app/(site)/[locale]/products/page.tsx`; `docs/seo-launch-2026-09-29.md`.

Sitemap and structured data

The dynamic sitemap reads the actual public CMS projection. It includes both languages for home, About, Contact, Resources, the two industries, all catalogue pages, nonempty category pages and each published product. Entries include reciprocal locale alternatives; product entries include the real public product-image URL. It does not invent modification dates. Drafts, empty categories, internal SKUs, basket/RFQ/verification, private routes and query-filter combinations are excluded.

The shared layout emits linked `Website` and `Organization` entities for EL AMAL, with stable `#website/#organization` IDs, URL and languages. It does not invent address, phone, hours, local-office claims, social identities or certifications.

Page family	Additional JSON-LD
Homepage	Shared Website and Organization graph
Catalogue/category	CollectionPage linked to the website; ItemList containing only displayed records; BreadcrumbList
Product in CMS mode	WebPage linked to website and Product; BreadcrumbList; Product with name, description, model, URL, actual image and recorded manufacturer
Demo product	WebPage and BreadcrumbList only; no real Product assertion
About	AboutPage plus BreadcrumbList
Contact	ContactPage plus BreadcrumbList
Resources	WebPage plus BreadcrumbList

Page family	Additional JSON-LD
Industry	Localized WebPage in addition to the shared site graph

An ItemList's numberOfItems is the current displayed-page count, while positions continue from that page's real offset. Product schema links page and product identities. The current code does not serialize the technical specification table as additionalProperty; the details are readable HTML. No offer, made-up zero price, AggregateRating, fake review, certification or authorized-distributor relationship is added. Manufacturer identity means who makes the instrument, not proof of EL AMAL's commercial authorization.

The WIKA relationship block appears on home/About only after approved English wording, Arabic wording and a safe HTTPS evidence URL are all supplied. It is otherwise absent. Ordinary search discoverability is the current goal; the presence of Product JSON-LD alone does not establish eligibility for a Google product rich result.

Source: src/lib/seo-discovery.ts; src/lib/product-schema.ts; src/components/site-schema.tsx; src/components/wika-evidence.tsx; src/app/(site)/[locale]/[information]/page.tsx; src/app/(site)/[locale]/industries/[slug]/page.tsx.

What the final SEO audit actually proves

The saved final read-only crawl began on 29 September at 14:46 UTC. It records 342 public HTML pages, 342 sitemap URLs, 155 checked asset URLs and zero detected issues under its implemented rules. All 151 product pages were reached in each language. The 155 assets are 151 product PNGs, /social-image, the icon URL, the versioned favicon URL and the plain favicon URL; they are not 155 different product photographs.

scripts/audit-live-seo.mjs starts from the sitemap/homepages, follows allowed public route families, and checks HTTP behavior, canonicals, reciprocal alternatives, language/direction, titles/descriptions/H1, robots, sharing metadata and parseable JSON-LD. It checks referenced social/product/icon assets, private-route discovery exclusions and missing-route handling. A configured page cap reached prematurely is a failure; it does not silently label a partial crawl complete. The final record has no detected issues, not a guaranteed absence of every conceivable SEO problem.

It does not prove Google indexed the pages, ranking, traffic, rich results, full content accuracy, external technical-PDF availability at the later crawl date, accessibility, field Core Web Vitals, every structured-data recommendation or whole-site security. The original PDF identity verification is a separate catalogue evidence set. Search Console property verification/submission and representative URL inspections remain owner follow-through. If the final public domain changes, matching old-to-new redirects, SITE_URL, redeployment, full recrawl and replacement sitemap submission are needed.

Source: artifacts/2026-09-29/completion/seo-final.json; scripts/audit-live-seo.mjs; docs/seo-launch-2026-09-29.md; catalogue/2026-09-27/datasheet-links.json.

8 Performance changes and measured limits

Changes that reduced real work

The interface uses Next.js 16.3.5, React 19.2.8, Tailwind 4.3.3, TypeScript 5.9.3 and Motion 12.43.0, with Payload 3.90.2 for CMS. Most catalogue content is rendered on the server; interactive code is concentrated in navigation, basket/forms, introduction and the conditionally loaded desktop instrument.

Four material improvements were made against the actual CMS catalogue:

1. **Main image discovery.** The product detail photo changed from lazy loading to eager/high priority. A prominent initial image should not wait as if it were below the fold. Cards remain lazy.
2. **Smaller hydrated data.** Basket consumers receive only ID/model/name instead of every technical row. Recorded uncompressed product HTML fell from 140,383 to 60,600 bytes after related payload changes. This is one sampled response, not a universal page-size constant.
3. **No public theme-negotiation restart.** Payload's `Critical-CH` header was scoped to admin routes so supporting browsers do not restart public navigation for CMS theme negotiation.
4. **Font and catalogue-read work.** Newsreader requests only the regular 400 weight actually used. Noto Sans Arabic is one variable face rather than repeated weight declarations, with unconditional shared preload disabled. Independent product/category pagination starts concurrently, while each collection reads one page at a time.

All public fonts still use `next/font/google, display: swap` and bundled self-hosted output. A clean build needs font retrieval; the visitor does not make a runtime Google Fonts CSS request. The earlier raw font-file calculation was 248,880 bytes; the expected optimized English raw-file total was 47,080, a saving of 201,800 bytes. A fresh deployed English browser observation measured 47,680 transferred bytes, including transfer overhead. These values describe slightly different measurements and should not be presented as a discrepancy or an exact timing improvement.

Arabic still loads its required font faces when CSS matches Arabic content. Removing shared preload is not evidence of zero Arabic font cost or final Arabic performance acceptance. Current final Lighthouse samples are English; Arabic received rendering/no-overflow/no-script checks, not a documented matching final Arabic Lighthouse run.

Catalogue reads use request-scoped React cache, not a persistent cross-request catalogue cache. A new request sees publication changes. Product and category streams run together with total concurrency at most two; pagination within each stream remains sequential, preserves order and reads every page. Failure of either stream rejects the load; no partial plausible catalogue or demo fallback is returned. Five targeted regressions cover concurrent start, bounded pagination, both failure paths and later-request freshness.

Source: `package.json; src/app/(site)/[locale]/layout.tsx; src/components/product-image.tsx; next.config.mjs; src/lib/load-catalogue.ts; src/lib/read-catalogue-records.ts; tests/read-catalogue-records.test.mjs; docs/performance-completion.md; PROGRESS.md.`

Current measured results

Saved mobile lab sample	Score	LCP	Total Blocking Time	CLS	Accessibility / Best Practices / SEO
English homepage, preceding font release, 14:36 UTC	91	2.890s	205.5ms	0	100 / 100 / 100
English actual product /en/products/cms-120, 14:38 UTC	90	2.919s	232ms	0	100 / 100 / 100
English homepage, final concurrent-read release, 14:50 UTC	89	2.758s	237.5ms	0	100 / 100 / 100

For normal communication these round to homepage 89-91, final LCP 2.8s/TBT 240ms; product 90, LCP 2.9s/TBT 230ms. The final homepage result is 89, not 91. The product's most recent saved 90 is the preceding font-release sample; there is no separate final-product Lighthouse file after the loader change. A lower LCP in one run does not mean every metric or overall score improved.

Earlier actual-product performance was 57/LCP 5.8s/TBT 590ms before the image/payload work, then 84/LCP 3.4s/TBT 160ms. An earlier actual-home sample was 78/LCP 4.0s. The demo-era homepage 100/LCP 1.2s came from different content and conditions and must not be reused as the real catalogue's current result.

The intended steady 90+ performance bar is not yet demonstrated consistently. LCP below 2.5s remains unmet in the current samples. CLS was zero in those samples, but this does not prove all real sessions have zero shift. Total Blocking Time is a lab measure and must not be relabelled INP. No field INP or real-user 75th-percentile LCP/INP/CLS evidence is recorded. These local simulated mobile measurements are not a full field Core Web Vitals assessment.

Remaining performance work includes representative repeated EN/AR home/catalogue/detail measurements, Arabic font-discovery/shift checks, investigation of server/catalogue readiness and render-blocking work, and a streaming strategy that preserves no-script content if pursued. None should be claimed complete simply because a cloud build succeeded or one score reached 90.

Source: artifacts/2026-09-29/completion/lighthouse-home.json; lighthouse-product.json; lighthouse-final-home.json; artifacts/2026-09-29/completion/final-browser.log; docs/performance-completion.md; PROGRESS.md.

9 Accessibility and verification boundaries

Implemented accessibility measures include a skip link to the main content, landmark structure, real links/forms/buttons, visible focus, menu expanded-state relationships, localized labels/status/error messages, controlled focus in the basket form, text-based availability labels, image alternative text, RTL-aware layout, no-script catalogue reading and reduced-motion alternatives. Decorative gauge layers and arrows are hidden from assistive technology where appropriate. Phone/contact/footer links and main controls have deliberate touch target heights; controls generally use 44-54px minimum interactive heights.

Reduced-motion CSS turns off transitions/animation and smooth scrolling, hides the introduction/replay, resets hero transforms and removes the moving route needle. The hero component also avoids importing its Motion behavior when the live media preference requests reduced motion. No-JavaScript styles hide the introduction and expose the mobile nav links, while server-rendered product content and native search/pagination remain readable. This is progressive enhancement for content, not an assertion that interactive RFQ/basket submission works without JavaScript.

The recorded launch browser audit covered ten representative public/login views: English home (1440px), Arabic home (390px), English catalogue (1440px), Arabic catalogue page 2 (390px), English actual product (390px), Arabic actual product (320px), English RFQ (390px), Arabic basket (390px), English Contact (390px), and admin login (1440px). It also sampled the authenticated stock page. Automated axe results found zero violations in the selected rules after the public-notice/login landmark fixes. Manual-review items remained. Its configured public rule tags were WCAG 2 A/AA, WCAG 2.1 A/AA and best-practice checks, so it is not evidence of comprehensive WCAG 2.2 conformance.

The same release evidence records the reduced-motion entrance as hidden, Skip to content as first keyboard focus, and an actual Arabic product readable with JavaScript disabled. Post-payload browser checks exercised basket add/quantity/reload/Arabic switch/remove, model search and menu Escape/focus restoration, then cleaned the test selection. The final completion browser log again records EN/AR home/detail samples without overflow or detected axe violations and an Arabic no-script product heading. Final private reports were also checked at 390px and 1440px, separately from the public route coverage.

A Lighthouse accessibility score of 100 is a result for the audited rules/sample, not an accessibility certification. Screen-reader task completion, high zoom/reflow, browser/device combinations, user testing, enabled real-submission error/recovery flows and final Arabic editorial review remain broader acceptance work. The originally requested Impeccable skill/suite was unavailable in the recorded project setup, so equivalent manual/source/browser work must not be labelled as actual execution of /impeccable /audit, /colorize, /typeset, /animate OR /polish.

Source: src/components/header.tsx; src/components/enquiry-preview.tsx; src/components/site-intro.tsx; src/app/(site)/[locale]/styles.css; artifacts/2026-09-29/launch/browser-audit.json; artifacts/2026-09-29/launch/check-browser.mjs; artifacts/2026-09-29/completion/final-browser.log; docs/kickoff-2026-09-17.md; docs/verification-2026-09-17.md.

10 Analytics reporting and remaining acceptance

Analytics is not the same as the demand report

The kickoff's ANA01 required product events, server-confirmed leads and a protected demand CSV, with controlled event reconciliation and no contact data in analytics. The existing source/package/scripts/tests contain no identified Google Analytics/gtag/dataLayer integration, browser event collector or product-view/search/datasheet-click/basket-add analytics instrumentation. There is no recorded analytics-provider configuration or event-reconciliation acceptance evidence. Those behavioral events must remain explicitly unimplemented/unverified rather than being inferred from Vercel hosting, server logs or an SEO crawl.

The private demand report is implemented. Owner/sales users can see aggregated saved enquiries, requested units and genuinely verified customer enquiries, with protected CSV export. Verified counts require stored customer verification status and timestamp, and exclude staff/demo verification from the default customer cohort. These are database-derived demand counts, not an external server-confirmed

lead event feed, conversion attribution, page-view analytics, sales revenue or an inventory-shortage report. The distinction preserves an honest status for the original ANA01 requirement: its reporting/export portion exists; behavioral/event collection and reconciliation are not evidenced as complete.

Source: docs/kickoff-2026-09-17.md ANA01/T11; docs/demand-reporting.md; src/lib/demand-service.ts; src/lib/demand-report.ts; src/app/(staff)/staff/reports/page.tsx; package.json; source search performed for this handbook.

Work and inputs still outstanding

Area	Remaining item	Why it matters
Client identity/content	Final logo, approved company/factory photography, accurate business facts and final About/Contact copy	Present content emphasizes the process and does not establish missing company history/address/hours/services
Direct contact	Approved international phone and WhatsApp numbers	Conditional links remain absent until valid values exist
WIKI relationship	Exact bilingual wording plus official evidence/certificate/listing	No distributor/authorization claim should be invented
Catalogue acceptance	Client technical and Arabic review; exact offered variants where relevant	Family data and source-link checks do not certify each offered configuration
Real inventory	Exact SKU definitions, opening quantities and accepted reservation/expiry/freshness policy	Dated public 90/61 model labels cannot become counted stock
Customer delivery	Owned sender/domain, sending service, frequent scheduler authorization and actual delivery/recovery tests	Receiving inbox is already known; it alone cannot send confirmation mail
Attachments	PDF/Excel and larger-file private quarantine, malware scanning/storage plus acceptance	Current photo subset does not fulfill the original 10 MB multi-format proposal
Policies	Approved privacy/retention/legal wording and appropriate public policy pages	There are no current dedicated public policy routes
Analytics	Agreed privacy-safe product/search/download/basket events, confirmed-lead event design and reconciliation if retained	Protected reporting alone does not complete original behavioral analytics scope
Search	Final domain decision, Search Console ownership/sitemap submission and actual index observations	Technical indexability is not a ranking or indexing result
Performance	Consistent 90+, LCP target and field/Arabic measurements	Current final homepage is 89; LCP remains 2.8-2.9s in the cited samples

Area	Remaining item	Why it matters
Accessibility/UX	Broader screen-reader, zoom, device, real form and final client walkthrough	Sampled automated checks cover only a subset of users/tasks
Operations	Automatic offsite backups and wider launch acceptance	A completed restore rehearsal does not schedule future backups
Monitoring	External monitoring deferred explicitly by Nour	Record as deferred; do not repeatedly request activation
Final acceptance	Nour's visual review and the client's operational/content launch decision	Agent measurements and source review are not owner approval

No repeat request is needed for all 29 catalogue pages, the main-page stock scope or the receiving inbox: those inputs are already settled. The next phase should collect the still-missing inputs together, preserve the approved navy/white/orange direction, and resume from the existing application rather than rebuilding the catalogue.

Source: docs/final-client-inputs.md; docs/phase-status.md; PROGRESS.md current milestone; docs/customer-intake.md; docs/private-enquiry-photos.md; docs/database-recovery.md.

11 Reading historical documentation correctly

Several historical documents retain their original checkpoint statements. Their dates and later amendments matter:

- docs/catalogue-publication-2026-09-28.md and older sections of the catalogue README still say indexing is disabled, show an old favicon 404, and list operational features as future work. The 29 September code/final audit supersedes those statements; indexing is enabled for eligible public pages and favicon URLs returned 200.
- artifacts/2026-09-29/completion/performance-report.md is the font investigation snapshot and says no new Lighthouse score exists yet. docs/performance-completion.md and the later Lighthouse JSONs record actual deployment measurements. Use the final 89 homepage result, with the observed 89-91 range.
- Older progress sections cite 34, 53, 58, 65, 121, 122 or 137 tests and 70%/85% delivery estimates. The current source checkpoint records 142 unit checks and a roughly 90% full-scope estimate; neither is a quality, ranking or security guarantee.
- The original four-industry proposal is not the same as the later two-industry client implementation. The original 250-model allowance is not an unfulfilled claim that 250 real source models were received.
- Existing “verified catalogue” or “reviewed catalogue record” labels describe the application's publication checks, not final client acceptance of Arabic, suitability, exact variants or stock promises.
- The 159 verified document identities, 150 linked entries, 164 document references and 160 unique URLs measure different things. They should not be collapsed into one figure.

Evidence priority: current source and final deployment evidence establish implemented behavior; the newest explicit user/client scope changes establish requested scope; original briefs establish remaining acceptance requirements where not superseded; older milestone prose is history.

Appendix A Published product route index

All 151 English/Arabic product route pairs and their model-group labels are listed in the companion catalogue route index, extracted from the final saved crawl. This keeps the main handbook readable while retaining every published product URL. See section 2 for all other public route families and exact pagination ranges.

Appendix B Source and evidence index

Topic	Primary implementation/evidence
Current release authority	PROGRESS.md current milestone; application 7f73267; artifacts/2026-09-29/completion/final-deployment.json
Initial and later scope	docs/kickoff-2026-09-17.md; docs/client-requirements-2026-09-24.md; docs/final-client-inputs.md
Public page templates	src/app/(site)/[locale]/page.tsx; products/page.tsx; products/[slug]/page.tsx; categories/[slug]/page.tsx; industries/[slug]/page.tsx; [information]/page.tsx
Locale shell/navigation	src/app/(site)/[locale]/layout.tsx; src/components/header.tsx; src/components/business-contact.tsx
Design and motion	src/app/(site)/[locale]/styles.css; src/components/hero-instrument.tsx; interactive-instrument.tsx; site-intro.tsx; instrument.tsx
Catalogue/publication	catalogue/2026-09-27/publication.json; datasheet-links.json; source-manifest.json; docs/catalogue-publication-2026-09-28.md
Public data boundary	src/lib/load-catalogue.ts; read-catalogue-records.ts; public-catalogue.ts; catalogue-details.ts; access.ts
Search and result UX	src/lib/catalogue.ts; src/components/catalogue-view.tsx; src/content/product-options.ts
Basket and RFQ	src/lib/basket.ts; src/components/basket-provider.tsx; quote-basket.tsx; enquiry-preview.tsx; direct-rfq.tsx; customer-enquiry-submit.tsx
Verification/photo UI	src/components/verify-enquiry.tsx; enquiry-photo-upload.tsx; docs/private-enquiry-photos.md
Metadata and SEO policy	src/lib/page-metadata.ts; site-policy.mjs; seo-discovery.ts; product-schema.ts; src/components/site-schema.tsx; next.config.mjs
Discovery endpoints	src/app/sitemap.ts; robots.ts; social-image/route.tsx; icon.svg; favicon.ico
Full SEO crawl	scripts/audit-live-seo.mjs; artifacts/2026-09-29/completion/seo-final.json; docs/seo-launch-2026-09-29.md

Topic	Primary implementation/evidence
Final performance	artifacts/2026-09-29/completion/lighthouse-final-home.json; lighthouse-home.json; lighthouse-product.json; docs/performance-completion.md
Browser/accessibility scope	artifacts/2026-09-29/launch/browser-audit.json; check-browser.mjs; artifacts/2026-09-29/completion/final-browser.log
Reporting versus analytics	docs/demand-reporting.md; src/lib/demand-report.ts; demand-service.ts; docs/kickoff-2026-09-17.md ANA01/T11
Remaining acceptance	docs/phase-status.md; docs/final-client-inputs.md; PROGRESS.md current milestone

Artifact files are local saved evidence and are ignored by Git. Preserve them through the project's private backup process when handing the project to a new machine. A source checkout alone may not contain them.

EL AMAL administration and staff operations handbook

Evidence date: 29 September 2026. These chapters describe the checked-in application at documentation checkpoint `f751ef5ee4f14378af3c1868936bf012b8dabb8f`, whose progress record identifies application release `7f7326704305012e9637702b3112b5c85f554833` as live. Procedures were checked against collection definitions, staff components, service logic and the installed Payload implementation. This documentation task did not change application records or perform the examples in production. The field and action names below are source-verified English labels; they are not a new browser walkthrough or visual approval.

1 What is ready to use and what remains inactive

Administration is at [EL AMAL admin](#). The live catalogue contains 151 model or model-group entries. Its dated folder-derived labels are 90 in stock and 61 out of stock, reported on 27 September 2026. Those figures describe catalogue families. They are not counted warehouse units and are not 151 orderable configurations.

The latest recorded production check found **zero exact SKU definitions**. Until the owner defines real configurations and a warehouse/owner records documented opening receipts, an empty stock screen is the expected result. It does not mean the company has no physical stock. No one should convert the 90 catalogue labels into quantities.

Catalogue management, private stock operations, saved-enquiry administration and demand reporting are implemented. Public customer submission, real customer email sending and staff password-recovery delivery remain disabled pending activation prerequisites. A verified sending identity/provider and frequent mail worker still need to be established; the existing daily maintenance run is not a frequent email scheduler. Do not promise that **Forgot password?**, submission, verification or notification emails work today. External monitoring was explicitly deferred by Nour.

Staff can read existing saved data where their roles allow. A blank report is legitimate when no matching enquiries exist. Ordinary staff cannot manually create a customer enquiry or mark one verified through the admin panel. This protects customer history and email-verification evidence.

The public catalogue image workflow and the private customer-photo workflow are separate. There is no general **Media** library in the current Payload configuration. Product photos live in deployed website files; private enquiry photos are encrypted database records.

2 Finding the right screen

Use the same production host for every route below. On a local development server, the paths are the same but the database may be different. Never use a development account or development data as evidence about production.

Purpose	Exact route	Navigation or screen text
Sign in	/admin/login	Email, Password, Login
Admin dashboard	/admin	Dashboard links include Open stock control and Open demand reports
Products	/admin/collections/products	Product documents are identified by Model
Categories	/admin/collections/categories	Category documents are identified by Key
Exact SKU definitions	/admin/collections/skus	Staff header link: SKU definitions; records identified by Sku Code
Saved enquiries	/admin/collections/enquiries	Staff header: Enquiries; records identified by Reference
Staff accounts	/admin/collections/staff	Records identified by email
Stock actions	/staff/inventory	Stock control
Demand report	/staff/reports	Page heading: What customers request.
Immutable hold records	/admin/collections/inventory-reservations	Inventory Reservations, within Stock control
Immutable movement records	/admin/collections/inventory-movements	Inventory Movements, within Stock control
Staff notification state	/admin/collections/notifications	Notification queue
Customer confirmation mail state	/admin/collections/verification-emails	Verification email queue
Owner maintenance summary	/admin/collections/delivery-operations	Delivery operations

Purpose	Exact route	Navigation or screen text
Private customer photos	/admin/collections/enquiry-attachments	Enquiry photos; list-cell action: Download photo

Payload uses /admin/collections/<collection>/create for a new document and /admin/collections/<collection>/<id> for an existing one. Use **Create New** only where the role permits creation. For stock, create entries through **Stock control**, never through either inventory collection.

Most collection and field labels are generated from their code names. For example, the actual generated labels are **Sku Code**, **External Id** and **Datasheet Url**, rather than acronym-normalized wording. The skus collection has no explicit plural label; the installed Payload formatter generates the awkward **Skuses**. The custom **SKU definitions** navigation link and /admin/collections/skus route are unambiguous. English and Arabic product values appear together under **English** and **Arabic** fields; this is not a separate language-selector workflow.

The staff header shows links for stock, reports, SKU definitions and enquiries without hiding every inaccessible destination. A visible link does not grant permission. The destination and its API enforce access.

Sign-in procedure

1. Open the production /admin/login page and use the account assigned to you.
2. Enter **Email** and **Password**, then choose **Login**. Do not post credentials into a report, chat, ticket or Git file.
3. Open the needed collection or the dashboard's stock/report link.
4. When finished on a shared computer, use the account's logout control. Closing a browser tab is not proof of logout.

The configured login lock is five unsuccessful attempts followed by a ten-minute lock window. Authentication token expiration is two hours; session refresh can affect what the user experiences. These are source configuration values, not a guarantee that every session lasts exactly two hours. If locked out, stop repeated guessing, check the correct environment and ask the owner for account help. Email recovery is currently unavailable.

3 Roles who can actually do what

Roles are stored as the exact values owner, catalogue-editor, sales and warehouse. Owner is the most privileged account. Assign staff by responsibility rather than sharing the owner login.

Operation	Owner	Catalogue editor	Sales	Warehouse
Enter administration	Yes	Yes	Yes	Yes
Read products, categories and SKU definitions	Yes	Yes	Yes	Yes
Create/edit categories and product drafts	Yes	Yes	No	No

Operation	Owner	Catalogue editor	Sales	Warehouse
Publish product content or save a change that remains published	Yes, with required review evidence	No	No	No
Unpublish a product through permitted update	Yes	Yes	No	No
Set product reviewer, review date and rights confirmation	Yes	No	No	No
Create an exact SKU or change its active flag	Yes	No	No	No
Change an existing SKU's identity	No; create another SKU	No	No	No
Read stock balances, reservations and stock ledger	Yes	No	Yes	Yes
Record receipt or adjustment	Yes	No	No	Yes
Create or release a hold	Yes	No	Yes	No
Dispatch a hold	Yes	No	No	Yes
Block/unblock units or confirm a physical count	Yes	No	No	Yes
Reconcile ledger	Yes	No	Yes	Yes
Read enquiries; edit workflow status/internal notes	Yes	No	Yes	No
Read notification and verification-email queues	Yes	No	Yes	No
View demand report/download its CSV	Yes	No	Yes	No
Read/download eligible private enquiry photos	Yes	No	Yes	No
Read Delivery operations collection	Yes	No	No	No
Create/update staff accounts	Yes	No	No	No
Read staff accounts	All	Own record	Own record	Own record

Normal admin/API deletion is denied for staff, categories, products, SKUs and enquiries. Stock history and customer-photo records also have no ordinary create/update/delete path. Hidden verification and request-limit collections are not available for staff administration, including the owner.

Permission details that matter: catalogue editors can read SKU definitions but cannot read warehouse quantities. Warehouse staff can see hold references and SKU snapshots in stock control but do not receive customer names, emails or the enquiry selector. A stock/report service rechecks the user's persisted staff role inside its database transaction; an outdated browser view is not authority to perform an action.

The ordinary staff create/update rule is owner-only, but Payload's separate default account-**unlock** permission is broader: this project does not override its default check for an authenticated user in the admin staff collection. Do not describe account unlocking as owner-only or treat this matrix as an independent security certification. An owner should handle lockout incidents operationally until that separate permission receives explicit review.

4 Maintaining categories

A category is a public grouping such as pressure, temperature or accessories. It organizes models; it contains no inventory count.

Field label	Meaning and requirement
Key	Required unique category identifier used in public category URLs and filtering
Name → English, Arabic	Required names for both languages
Description → English, Arabic	Required explanatory copy for both languages

1. As owner or catalogue editor, open `/admin/collections/categories`.
2. Open the existing category by **Key**. Use **Create New** only when a genuinely new category is needed.
3. For a new category, agree a stable, readable URL-safe **Key**. The field is unique text; the application does not generate a safe slug for you.
4. Fill both **Name** values and both **Description** values. Keep technical meaning aligned across languages.
5. Choose **Save**. Categories do not have a product-style draft/publication workflow.
6. Check the public category page in both languages after the save. A category appears in the public catalogue only when it has at least one eligible published product.

Changing a category's name/description can affect published pages immediately. Changing **Key** changes the public grouping identifier and can break previously shared category links; no automatic redirect is configured in this collection. There is no category delete button authorized by the application. When retiring a grouping, review product assignments and URL consequences with the owner rather than assuming deletion or hiding is available.

Implementation: Categories in `src/cms/collections.ts` controls editing. `toPublicCatalogue` in `src/lib/public-catalogue.ts` includes only categories used by eligible published products. This prevents empty navigation groups from appearing as if they contain catalogue stock.

5 Product catalogue draft review publish and correct

A **Product** is a catalogue model/family. A **SKU** is a specific orderable configuration of that model. The product's public page explains the family and its supported options; its availability label does not reserve a unit or promise every configuration is stocked.

Product fields

Exact label	What to enter; who controls it
Catalogue Details	Structured JSON for reviewed manufacturer image, bilingual technical rows, dated family availability and datasheets. See the next section.
External Id	Required unique stable catalogue/import identifier. Retain an existing identifier when correcting content; do not manufacture a second record for the same imported entry.
Model	Required manufacturer model or printed model-group label. Preserve the real designation.
Category	Required relationship to an existing category.
Name → English, Arabic	Required human-readable model name in both languages.
Description → English, Arabic	Required technical introduction in both languages.
Instrument Type	Optional filter value: Pressure gauges, Pressure transmitters, Pressure switches, Temperature instruments, or Valves & accessories.
Applications	Optional multiple selections: Oil & gas, General industry. Choose only applications supported by manufacturer evidence.
Datasheet Url	Optional HTTPS link to the actual manufacturer document. The URL validator does not prove that the PDF is correct or licensed.
Source Ref	Required evidence pointer for the submitted facts and asset source. Visible to authenticated staff, excluded from normal anonymous field access.
Reviewed By	Owner-set relationship to the staff reviewer. Readable in normal field access by owner/catalogue editor.
Reviewed At	Owner-set review date. Publication requires a parseable date.
Rights Confirmed	Owner-set checkbox confirming rights have been reviewed; defaults false.

Review fields record an operator decision. Their presence is not proof that a manufacturer or legal reviewer approved the content. The application verifies required values; a person must verify their truth.

Creating or updating a draft

1. Open `/admin/collections/products`. Search for the **Model** first so an existing family is not duplicated.
2. Open the record to correct it, or choose **Create New** for a new approved catalogue entry.
3. Enter the stable **External Id**, exact **Model**, **Category**, both **Name** values and both **Description** values.

4. Choose evidence-supported **Instrument Type** and **Applications**. Preserve manufacturer codes, units, signs, ranges and configuration caveats when translating.
5. Add **Source Ref**, verified document links and the reviewed **Catalogue Details** content. If an image file is new, use the file-preparation process in section 6 first.
6. Choose **Save Draft**. Payload's draft action can save incomplete required fields, but invalid non-null Catalogue Details still fails its custom validation. A successful draft save is not a successful publication review.
7. Record what the owner still needs to verify. There is no implemented editorial approval inbox or automatic review email; arrange review through the team's existing process.

Saving draft edits to a product that already has a published version does not replace its public version. The public loader reads `draft:false`. Draft and public content can therefore differ deliberately while a correction is being reviewed.

Publishing reviewed content

1. The owner opens the draft and reviews English and Arabic content, model identity, images, technical rows, dates, links and asset rights against the evidence.
2. Set **Reviewed By**, **Reviewed At** and **Rights Confirmed** accurately.
3. Confirm that both language names/descriptions and **Source Ref** are filled. These are explicit publication requirements.
4. Choose **Publish** or **Publish changes**, as appropriate to the document state.
5. Check the actual public product page in both English and Arabic, including the image, specification rows, availability date and datasheet links. The public detail route uses `/en/products/cms-<numeric product id>` OR `/ar/products/cms-<numeric product id>`; it does not use the typed model name as the slug.
6. Treat Nour's review as visual acceptance. A successful save, valid JSON or correct HTTP response does not provide visual acceptance on his behalf.

Two independent gates apply: the product save hook restricts publication to owners with required review evidence, and the public projection excludes unpublished/unreviewed records. A fresh public request reads current published records without substituting demonstration data if the real catalogue fails.

Removing or correcting public content

For a factual correction, edit and save a draft, then repeat owner review and publication. For immediate withdrawal, **Unpublish** changes the product to draft and removes it from the eligible public catalogue. The code permits owner or catalogue editor updates and blocks only the published state for nonowners, so catalogue editors can unpublish; this is a current permission, not an owner-only rule.

Products cannot be deleted through normal admin access. Unpublishing does not delete an old enquiry snapshot, rewrite demand history, release a stock hold or deactivate an SKU. If any of those business actions are needed, handle them deliberately in their own screens.

6 Product photos specifications datasheets and availability

What the Catalogue Details editor accepts

Catalogue Details is a JSON editor, not a media picker or separate specification form. Preserve its structure and obtain technical help if JSON editing is unfamiliar. It accepts these fields:

JSON path	Contract
manufacturer	Must be exactly WIKA; another manufacturer's content is not supported by this schema yet.
image.src	Local /images/products/<filename> path. Allowed extensions: jpg, jpeg, png, webp, avif. No remote image URL here.
image.width, image.height	Whole pixel dimensions between 1 and 10,000. These must describe the actual asset.
image.alt.en, image.alt.ar	Nonempty descriptive alternative text in both languages, maximum 300 characters each.
image.sourceUrl	Credential-free HTTPS evidence URL identifying the image source.
specifications	Array of up to 80 rows. Each row contains label.en, label.ar (up to 200 characters) and value.en, value.ar (up to 4,000 characters).
availability	Exactly in-stock, out-of-stock or check.
availabilityReportedAt	Real non-future date written YYYY-MM-DD. It dates the family-level report.
datasheets	Up to 20 entries, each containing a nonempty title (up to 200 characters) and a credential-free HTTPS url.

The parser normalizes the allowed fields and strips unknown keys. Do not put credentials, private customer details, prices, warehouse counts or experimental properties in this object. The schema permits empty specification/datasheet arrays and permits Catalogue Details to be absent; passing its validation therefore does not by itself prove a complete product page.

Adding or replacing a catalogue image

1. Confirm the exact model/family, source and permission to reuse the image. Use real approved manufacturer material for this catalogue.
2. Prepare a supported raster file with accurate dimensions and English/Arabic alternative text.
3. Have the technical maintainer add the asset to `public/images/products/` through the reviewed source/deployment process. This is not done through an admin **Upload** button, because no catalogue Media collection exists.
4. Verify the deployed `/images/products/<filename>` actually opens. A valid path string in JSON does not create the file or check that it exists.
5. Update `image` inside **Catalogue Details** in a product draft, preserving the other reviewed fields.
6. Follow owner review/publication, then inspect both public language pages.

This release intentionally keeps public manufacturer assets with website source files. It gives reproducible deployed assets but means nontechnical staff need a maintainer for new image files. Private enquiry photos cannot be copied into this role automatically; they are customer information with a separate access/retention policy.

Updating technical rows or datasheets

1. Open the source manufacturer material for the exact model and check its revision and applicability.
2. Update the English and Arabic specification rows together. Retain minus signs, decimal values, units and model-family qualifiers.
3. Set the correct HTTPS document URL and a clear title. The older top-level **Datasheet Url** and the **Catalogue Details** datasheets array both exist; inspect the resulting product page so links remain consistent.
4. Keep source evidence in **Source Ref** and image provenance in `image.sourceUrl` as appropriate.
5. Save a draft and obtain owner publication review. Opening a PDF and confirming the correct model is a content check beyond URL syntax validation.

Updating family-level availability

1. Obtain a dated business statement about that catalogue family/group.
2. In a product draft, update `availability` and `availabilityReportedAt` together.
3. Use `check` when confirmation is required; use a real date, never a future date or a rolling date that suggests a fresh warehouse count.
4. Review and publish through the owner workflow.

This does not alter SKU stock. Receiving, blocking, dispatching or confirming exact stock also does not update these public labels. A future policy that maps exact configurations to public family availability is still required before they can be linked safely.

7 Creating and maintaining exact SKU definitions

An SKU distinguishes the actual supplied configuration: for example a specific manufacturer part number, measuring range, connection and other order-defining attributes. Those attributes vary by instrument. A generic family name is insufficient where the warehouse holds several variants.

Field label	Meaning
Sku Code	Required unique internal SKU code
Product	Required link to the catalogue product/family
Manufacturer Part Number	Optional schema field; record the actual number when available
Configuration	Required JSON describing the exact orderable configuration
Active	Checkbox, default true; controls new receipts and holds

1. Owner: open **SKU definitions** or /admin/collections/skus.
2. Check whether the same configuration already exists.
3. Choose **Create New** and fill **Sku Code**, **Product**, known **Manufacturer Part Number** and **Configuration** using actual warehouse/purchasing evidence.
4. Review the values before saving. The code does not impose a measurement-specific schema inside Configuration or prove commercial completeness for you.
5. Save the definition. It starts with zero on-hand units. It does not copy the product's in-stock label into a count.
6. Use **Stock control** to record a documented receipt/opening quantity separately.

The SKU code, product relationship, manufacturer part number and configuration are immutable **from creation**, even before the first receipt. Correcting a mistaken identity means creating a new accurate SKU and deactivating the old definition. It is not safe to relabel existing units. A relationship object and its same numeric product ID are treated as the same identity; JSON object key order alone is not a changed configuration.

To retire an SKU, the owner opens it, clears **Active** and saves. Inactive SKUs remain in history. They cannot receive new stock or new holds, but existing holds can still be dispatched/released and documented adjustments, blocking/unblocking, counts and reconciliation remain possible. Deactivation does not automatically cancel a hold, remove stock or reassign units to the replacement SKU.

8 Working the enquiry inbox

Use /admin/collections/enquiries as owner or sales. The default list columns are **Reference**, **Company**, **Status**, **Source** and **Created At**. Opening a record shows submitted customer details and item snapshots. This is private customer information; keep exports/screenshots within the authorized business workflow.

Read-only submitted facts

Reference, **Locale**, **Source**, **Name**, **Email**, **Company**, **Notes**, **Items**, **Verification Status**, **Verified At** and **Delivery Status** are immutable through ordinary admin updates. Inside **Items**, each submitted line retains **Product Id**, **Model**, **Name En**, **Name Ar**, **Quantity** and **Range**. Submitted quantities are whole numbers from 1 to 9,999; an enquiry can contain up to 100 lines. The hidden request/fingerprint fields support duplicate protection and are not everyday staff controls.

Changing the current product catalogue does not rewrite these snapshots. That is intentional: staff and reports need to know what the customer actually asked at submission time.

The two editable fields

Field	Values/use
Status	new, reviewing, awaiting-customer, quoted, closed
Internal Notes	Staff's operational notes, up to 10,000 characters

The statuses are workflow labels. The code allows a permitted staff member to select any listed status; it does not enforce a staged sales approval sequence. Setting **quoted** does not generate a price, quotation PDF or email. Setting **closed** does not release holds or erase demand history.

Suggested staff procedure

1. Open the enquiry and verify **Source**, **Verification Status** and **Verified At** before treating it as a confirmed customer request.
2. `Source = cms`, `Verification Status = verified` and a saved **Verified At** timestamp are required for stock allocation. `test-verified` is a test state, not genuine customer verification; `demo-source` records remain demo even if other fields appear verified.
3. Read the requested model, quantity and range. Compare the request with the actual manufacturer's options and the customer's accompanying notes/photos.
4. Change **Status** to `reviewing` while investigating and add concise **Internal Notes** describing actions and unresolved facts.
5. If clarification is needed, use `awaiting-customer` and record what is missing. Communication outside the website remains a separate authorized business action; the status does not send it automatically.
6. Once a quotation has actually been prepared through the agreed business process, use `quoted`. For a finished request, use `closed` and record the outcome.
7. Review any related holds separately before closing. A new hold cannot be created against a closed enquiry, but existing hold release/dispatch operates on the hold and does not automatically stop when the enquiry closes.

There is no staff **Create New enquiry** workflow and no editable button that honestly converts an unverified enquiry to verified. If the customer's original immutable line is wrong, document the correction and arrange a correct new request through an authorized workflow once intake is active. Do not rewrite historical quantities to make a stock allocation fit.

Delivery state use the queue not the legacy field

The enquiry's **Delivery Status** currently has only the value `not-configured`, and staff cannot change it. This is a legacy field, not the current notification delivery ledger. Open **Notification queue** and inspect **Status**, **Attempts**, **Next Attempt At**, **Sent At**, **Provider Message Id** and **Last Error** for the matching **Reference**.

Notification statuses are `disabled`, `pending`, `processing`, `sent`, `failed`. Confirmation-mail queue statuses are `pending`, `processing`, `sent`, `failed`, `cancelled`. Both collections are read-only for owner/sales. A `sent` queue status means the provider accepted the message; it does not prove arrival in the recipient inbox or a completed sale. Do not reset attempts or manufacture a new date through database edits to force delivery.

The Enquiries collection currently contains the old description "Email delivery and stock reservation are not active yet." Its stock-reservation statement is stale. Stock control is implemented, while current production email/intake activation remains off for the separate prerequisites explained above.

Viewing a customer's technical photo

1. As owner/sales, open **Enquiry photos** at `/admin/collections/enquiry-attachments`.
2. Match **Enquiry** or **Reference** to the request being investigated. The default list shows filename, enquiry, byte count, expiry and download action.
3. Select **Download photo**. The protected endpoint downloads a file named `technical-photo.jpg` or `technical-photo.png`; the submitted filename remains metadata in the admin record.
4. Handle the downloaded file as private customer information. The application does not control retention of copies that staff download.

The implemented upload workflow accepts up to three reconstructed JPEG/PNG files per genuinely verified enquiry, each up to 2 MiB, and limits source decoding to eight million pixels. Live application access ends after 30 days. Expiry denies download immediately; physical cleanup depends on the retention worker. No PDF, Excel, SVG, video or general-document upload/scanning workflow is active. In the current disabled-intake state, do not promise a customer that they can complete verification/upload today.

9 Reading stock control before changing anything

Open `/staff/inventory` with an owner, sales or warehouse account. Begin in **Choose the exact SKU**:

1. Type a code or model in **Find a SKU or model**. The search refreshes automatically after a short delay.
2. Select the exact entry from **SKU**. Inactive entries are marked (*inactive*).
3. Expand **Review configuration for ...**. Confirm the configuration before allocating or recording units.
4. Read **On hand, Held, Blocked, Available** and the ledger-consistency message.
5. Review **Physical count review** and its last explicit confirmation time.
6. Use **Refresh balance** when another colleague may have made changes.

Balance	Plain-language meaning
On hand	Physical units represented by receipts, adjustments and dispatch movements
Held	Unconsumed units allocated by unexpired active reservations
Blocked	Physical units kept out of allocation, for example during inspection
Available	On hand minus active held units minus blocked units

Expired holds stop reducing **Available** immediately. An expiry ledger entry can follow later; the screen explains any pending expired units. Counting those units as available is intentional and does not mean they were dispatched.

The screen displays up to 100 matching SKUs, the latest 100 matching eligible enquiries, up to 100 reservations with active holds first, and the latest 100 ledger entries. Refine searches rather than assuming an item does not exist outside those bounds. Older reservations and movements are available in the corresponding read-only admin collections.

The stock view does not show a manufacturer part number as a separate visible value; it shows the code, linked model and Configuration JSON. If the part number is needed for a check, open the SKU definition

too. The saved hold snapshot contains code, product ID, model and configuration; it does not add a separate part-number field.

Rules shared by every stock action

In **Record a stock action**, choose the action under **Action**, complete its specific fields and fill **Reason or supporting document reference**. That reason is required, trimmed, nonempty and limited to 1,000 characters. Prefer concrete receiving/dispatch/count/inspection references that another staff member can trace.

Quantities are whole units. Ordinary action quantities are 1-999,999; adjustments may be negative but cannot be zero. Physical confirmation accepts 0-999,999,999. The maximum supported on-hand balance is 999,999,999. There is no fractional-unit, unit-conversion or valuation feature in this ledger.

The successful response states the saved action and its **Audit entry** number. Read the changed balance and the latest audit row after each action. A reasoned entry is permanent: correct mistakes with another authorized movement, not by editing/deleting old entries.

If an action times out

Do not repeatedly start new copies of the action. The screen retains the original command and unique request key in the current tab's session storage, tied to the staff account. **Confirm an unfinished action** shows **Retry original action**. Use that button without changing the retained details; an accepted repeat returns the original movement and says no duplicate was created.

A definite client/permission validation rejection clears the pending command so the form can be corrected. Network/server uncertainty retains it because the transaction might already have committed. If the tab/session storage was lost, check the audit trail before recreating the operation. A different actor or altered details cannot reuse the same accepted request key.

10 Stock procedures one action at a time

All figures in the examples below are invented training arithmetic. Do not enter them as EL AMAL's actual opening stock.

Record a receipt owner or warehouse

1. Confirm the exact physical SKU and receiving document; the SKU must be active.
2. Select **Record receipt** in **Action**.
3. Enter the number of units physically received in **Quantity**.
4. Fill **Reason or supporting document reference**, such as the actual supplier receipt reference and receiving explanation.
5. Choose **Record receipt** and verify the increased **On hand** and corresponding audit entry.

Example: receiving five units adds five to On hand and Available when no other action occurs. Creating an SKU alone does not do this. For initial loading, agree and document the physical opening count with the owner before recording receipts; do not invent supplier receipt numbers.

Record an adjustment owner or warehouse

1. Compare the actual counted quantity with the ledger and establish the cause of the discrepancy.
2. Select **Record adjustment**.
3. In **Signed change (use a negative number for a reduction)**, enter the difference, not the new total.
If the ledger shows ten and the supported physical total is eight, enter -2.
4. Record the actual count/correction evidence in the reason.
5. Choose **Record adjustment** and review the resulting balance and audit trail.

An adjustment cannot make the balance negative or consume units already held or blocked. It does not silently cancel commitments. If the count conflict affects committed units, resolve the underlying hold/block situation with authorized staff and preserve the reason. Adjusting does not count as **Confirm physical count**; confirm separately after the correct total is established.

Create a hold owner or sales

1. Confirm the exact SKU configuration and current available balance. Review the physical-count policy message first.
2. Select **Create hold** and enter **Quantity**.
3. Use **Find a verified enquiry** to search the actual **Enquiry reference**, then select it from **Verified enquiry**.
4. Select **Requested line**. The list shows model, submitted quantity and range where supplied.
5. Compare that range and the required connection/options to the SKU configuration yourself. The application checks the catalogue product identity, or case-insensitive exact model name for a customer-specified request; it does not understand range compatibility, substitute models or technical equivalence.
6. Choose the agreed **Hold expires (your local time)** value. It must be in the future and within 30 days. The browser converts this local time to UTC. Check the computer's timezone; this field is not automatically forced to Cairo time.
7. Add the reason/agreement reference and select **Create hold**.
8. Record the generated hold reference and check the **Reservations** table. The original quantity, expiry and configuration snapshot are retained.

A hold increases Held and decreases Available while leaving On hand unchanged. It requires a genuinely verified, open, CMS customer enquiry with a saved verification time. A test or demo record does not qualify. Across all SKUs allocated to the line, active held quantity plus already dispatched quantity cannot exceed the submitted line quantity.

The 30-day maximum is a technical bound, not a default commercial promise. There is no automatic 48-hour hold, automatic allocation or substitution. There is no edit-expiry button. If an expiry needs changing, review and release the remaining hold, then create a new hold with a new explicit expiry if the request and availability still permit it.

Release all or part of a hold owner or sales

1. Select the SKU, then **Release hold**.
2. Select the right **Active reservation**, checking its remaining units and enquiry reference.

3. In **Quantity to release**, enter the number actually being freed. It may be less than the remaining hold; it cannot exceed it.
4. Enter the reason, then select **Release hold**.
5. Verify Held decreased, Available increased and On hand stayed unchanged. Check the reservation's **Released** total.

A partial release leaves the remainder held until the original expiry. Releasing the final remaining unit closes the reservation. A released unit may be allocated again to the enquiry if total outstanding plus dispatched quantity permits it. Release is not a physical shipment or a stock write-off.

Dispatch all or part of a hold owner or warehouse

1. Confirm the actual physical shipment, exact configuration and dispatch document.
2. Select **Dispatch hold** and choose the correct **Active reservation**.
3. Enter only shipped units in **Quantity to dispatch**.
4. Include the dispatch document reference in **Reason or supporting document reference**.
5. Choose **Dispatch hold**, then verify On hand and Held both fell by the shipped amount and the reservation's **Dispatched** total increased.

Available usually stays unchanged for this operation because the dispatched units were already held. Remaining units stay held at the original expiry. Dispatching the final remaining unit closes the hold. After expiry, dispatch is rejected; review a new valid allocation before shipping rather than backdating the event.

Already dispatched quantities continue to count toward the original enquiry line after its remainder is released or expires. This prevents the same requested units being allocated twice. There is no standalone unreserved dispatch action, shipment reversal button, return authorization or credit-note module. A returned unit needs a documented stock process and an explicit decision about the customer's remaining requirement; do not assume a receipt reverses historical dispatch demand.

Block stock owner or warehouse

1. Confirm why specific available units must be unavailable for allocation, for example an actual inspection hold.
2. Select **Block stock** and enter **Quantity**.
3. Add the supporting reason/reference and choose **Block stock**.
4. Check Blocked increased and Available fell while On hand stayed unchanged.

You cannot block units that are already held or block more than Available. Blocking is an aggregate SKU balance, not a separate lot, serial-number or inspection-case record. Include enough detail in the reason to explain the business evidence.

Unblock stock owner or warehouse

1. Confirm the inspection/release decision.
2. Select **Unblock stock**, enter the quantity and record the evidence.
3. Choose **Unblock stock** and check Blocked decreased and Available increased. On hand stays unchanged.

The quantity cannot exceed the SKU's current blocked balance. Unblocking does not create units and does not create a customer hold.

Confirm a physical count owner or warehouse

1. Count **all** on-hand units for the exact SKU, including held and blocked units.
2. Compare the count with the displayed **On hand**. Investigate differences first; use a documented adjustment only when justified and permitted.
3. Select **Confirm physical count**.
4. Enter the full **Physical on-hand count**, including zero if zero is the verified count.
5. Enter the actual count/review reference and choose **Confirm physical count**.
6. Check the new confirmation timestamp under **Physical count review** and the audit row's **Physical count** value.

Confirmation records a review; it does not change balances. A mismatching count is rejected. Receipts, adjustments, dispatches and reconciliation do not pretend a full count took place.

Reconcile ledger owner sales or warehouse

1. Select the SKU and read the consistency message.
2. If totals agree, select **Reconcile ledger**, enter the reason and run it to record a review and materialize up to 100 due expiry entries for that SKU.
3. Review **Stock audit trail** and the resulting balance.

Reconciliation compares movement totals against immutable reservations and consumption events. It does not compare against an unentered warehouse count or silently fix corrupted data. If the screen says **Ledger totals disagree**, the action form is disabled, including reconciliation. The service also rejects mutations on inconsistent history. Ask the owner/technical maintainer to investigate; simply clicking Reconcile is not a repair mechanism.

Expired holds

An expired hold releases only its unconsumed remainder. Shipment and release totals already recorded stay intact. Availability stops counting the hold as soon as expiry is reached, even if the worker is delayed. The next stock action or maintenance worker can record its expiry event. Such events use actor role system, no staff actor and the fixed reason Hold expiry reached; availability released automatically.

The scheduler's expiry work is bounded and may skip a busy SKU. This does not extend the reservation's commercial validity. Neither an expired nor closed reservation can be selected for a new release/dispatch. A partially dispatched reservation can end in state **Expired** while its **Dispatched** column still correctly shows previously shipped units.

Optional freshness policy

The stock screen always shows the latest explicit count confirmation. An optional environment setting, INVENTORY_FRESHNESS_HOURS, can require a recent count before **Create hold**. It accepts a whole number from 1 to 8,760 hours. There is no default and no admin text box that sets this policy.

When unset, the screen says the interval has not been configured and permits manual-review allocation. When configured, unconfirmed/stale stock blocks new holds until the owner/warehouse confirms a

count. An invalid setting blocks new holds with a service error. Existing hold dispatch/release and corrective actions remain available. Exactly reaching the configured age is stale. The owner must choose the real operating interval before a maintainer configures it; do not infer a 48-hour policy from an old proposal.

Worked example understand the arithmetic

Assume a training SKU has twelve physically received units and a real qualifying enquiry asks for six:

Training action	On hand	Held	Blocked	Available
Receive 12	12	0	0	12
Block 2 for inspection	12	0	2	10
Hold 6 against the enquiry	12	6	2	4
Dispatch 2 from the hold	10	4	2	4
Release 1 from the hold	10	3	2	5
Remaining 3 expire	10	0	2	8
Unblock the inspected 2	10	0	0	10

The original hold remains six units: two dispatched, one released, three expired. The enquiry's already shipped quantity remains two. A physical count confirmation would enter ten, not Available at an earlier step. This is arithmetic illustration only, not a stock-loading instruction.

11 Demand reporting and CSV exports

Owners and sales open `/staff/reports` or **Open demand reports**. Warehouse and catalogue-editor accounts cannot access it. The report counts saved enquiry demand; it does not measure website visitors, sales revenue, stock shortages, quotation acceptance or purchases.

Run a report

1. Under **Choose the view**, set **From** and **Through**. Dates are inclusive Cairo calendar dates, including daylight-saving handling. The default is the last 30 days including today.
2. Choose **Enquiry group**: **Customer enquiries**, **Verified customers**, **Unverified customers**, **Test verification only**, **Demo only**, or **All groups, shown separately**.
3. Optionally enter **Model contains** and **Range contains**. These are literal case-insensitive substring filters on the displayed privacy-safe labels.
4. Choose **Groups per page** (normal choices: 25, 50 or 100) and select **Apply filters**.
5. Read **Saved demand** totals and **Demand by model and range**. Use **Previous page**, **Next page** or **Reset filters** as needed.

The date range may contain at most 93 calendar days, must begin in 2000 or later, and cannot end in the future. **Model contains** accepts up to 120 characters and **Range contains** up to 160; control characters are rejected.

Interpret the numbers correctly

Label	Meaning
Enquiries	Distinct matching saved submissions in the selected view
Requested units	Sum of quantities on matching submitted lines
Verified customer enquiries	Distinct matching customer submissions with verified state and a stored verification timestamp
Table Requests	Distinct enquiries containing that model/range/group combination
Table Units	Sum of matching line quantities in that group

Rows are sorted by requested units, highest first. An enquiry containing several different lines can appear in several rows. Add row Units when you mean units; do **not** sum row Requests and call the result distinct enquiries. Two matching lines from the same enquiry contribute their quantities but only one request to that same row.

Date filtering uses original submission time, while verification group uses the current saved verification state. Therefore a report for last week can change its verified/unverified split when a customer verifies later. Closed enquiries remain historical demand. A verified status without a verification timestamp belongs to unverified demand. test-verified is separate; demo-source records stay in the demo group. The default Customer enquiries view excludes test and demo records.

Why some labels are generalized

For submitted catalogue lines, model names come from the original snapshot, not the current product record. Direct customer requests can contain arbitrary identifying text, so their model is shown as **Customer-specified model**. Searching the customer's typed model will not recover that hidden raw value in the report.

Only a narrow numeric measurement expression is displayed as a range, for example 0-10 bar or -20 to 80 °c. Other free text becomes **Range requires staff review**; an empty range becomes **Range not specified**. The same range filtering applies to catalogue lines as well as direct requests. Similar expressions stay separate; the report does not convert psi to bar or merge equivalent units/ranges.

Open the private enquiry inbox when original details are needed. The report deliberately omits customer names, emails, company names, notes, attachments, request references and database IDs. It does not identify exact SKUs and cannot calculate a shortage from the public family availability labels.

Download CSV

1. Apply and review the intended date/group/model/range filters first.
2. Choose **Download CSV** beside **Demand by model and range**.
3. Open the downloaded file in a trusted spreadsheet application. Its filename follows
el-amal-demand-YYYY-MM-DD-to-YYYY-MM-DD.csv.
4. Check the period and filters before sharing. Even privacy-reduced aggregate demand remains private business information.

CSV exports every matching group across all pages, not just the visible page. Its exact columns are **From (Cairo), Through (Cairo), Model, Range, Enquiry group, Requests, Units**. It has a UTF-8 byte-order mark for Unicode/Arabic handling. Formula-leading text is escaped; a negative temperature range can intentionally begin with an apostrophe so a spreadsheet treats it as text.

The download button appears only for 1-2,000 matching groups. Above 2,000, narrow filters; the endpoint refuses an oversized export rather than truncating it. Independently, no report processes more than 20,000 source lines in its selected date period. That cap is checked before group/model/range filtering, so a shorter **date period** is needed when the source-line cap is exceeded. A no-data screen contains no demo/filler statistics.

Technical contract and rationale

GET `/api/staff/demand-report` accepts `start, end, cohort, model, range, page, pageSize` and `format`. The cohort values are `customers, verified, unverified, test, demo, all`; `format` is `json` or `csv`. API page size can be 1-100 and page 1-20,000. Duplicate/unknown parameters, invalid dates and oversized filter strings are rejected.

The service performs a read-only repeatable-read database transaction with a persisted role check. It reads immutable enquiry snapshots and never joins current catalogue names, writes stock or sends mail. JSON, CSV and error responses carry `private/no-store` and `noindex` protections. Its eight-second statement timeout and source/export limits bound work and prevent apparently complete totals from silently containing partial data.

12 Staff account administration and recovery limits

Add an account owner

1. Open `/admin/collections/staff` and check that the intended account does not already exist.
2. Choose **Create New**.
3. Enter **Email, Password**, the matching **Confirm Password** where the authentication form requests it, and **Role**.
4. Choose one of `owner, catalogue-editor, sales, warehouse`. The default is `catalogue-editor`; do not leave it by accident if the person's role is different.
5. Use a unique password/passphrase of 15-128 characters. Repeated-character and several common long passwords are rejected. A simple increase in length alone is not a reason to share or reuse a password.
6. Save, then arrange secure credential handover through the team's chosen private process. The application does not currently provide a functioning welcome/invitation/recovery-email onboarding flow.
7. Have the staff member sign in and check only the capabilities required for their job.

Change a role or password owner

Open the existing account, verify the email identity, change **Role** or use **Change Password**, complete the required password confirmation and save. Normal staff updates, including ordinary password/profile changes, are owner-only in the collection access rules. Do not promise that a non-owner can save an ordinary self-service profile/password edit just because an **Account** screen is visible.

Newly set passwords must satisfy the current 15-128 character rule. Existing older passwords were not automatically changed by that rule. The project record states that the existing owner password has not been rotated; this handbook does not reproduce it or certify its strength.

Avoid changing the last available owner's role away from owner. The source does not implement a last-owner safeguard. If all owner access is lost, a trusted technical recovery operation is required; the public first-register endpoint is deliberately blocked.

Removing access is not yet a normal disable switch

Staff records have no **Active**, **Disabled** or equivalent revocation field, and normal deletion is denied. Moving someone to catalogue editor still leaves administration and catalogue privileges; it is not a full revocation. A complete offboarding/session-revocation procedure needs an authorized technical operation and verification. Do not describe a role change as deleting an account or promise that every previous browser session was terminated.

Forgot password and future activation

Forgot password? may appear as part of Payload's authentication interface, but the application route currently returns **Staff recovery unavailable** unless all recovery configuration is ready. Do not instruct staff to expect a reset email today.

After explicit activation and verification, the implemented recovery design uses same-origin JSON requests, request limits, neutral account-existence responses, a 30-minute reset-token expiry, a ten-minute minimum reissue interval and a database lock preventing concurrent successful reuse of the same reset token. The dedicated staff adapter sends directly through the configured provider; it is distinct from the scheduled customer confirmation/notification queues. Production activation remains pending, and no real email-delivery outcome was demonstrated by writing this handbook.

13 Common problems and what the operator should do

Message or symptom	Meaning	Correct next step
Administration unavailable, Stock control unavailable	CMS gate/configuration or service unavailable	Ask the technical maintainer to check the environment; do not create a new production database or owner.
Stock/report page asks you to sign in after login	Wrong role, missing/expired session or wrong host	Confirm the host and account; sign in with the authorized role. A visible navigation link does not override access.
No matching SKUs	No definitions exist or search is too narrow	Production's last check had zero real SKU definitions. Owner must define genuine configurations; do not infer units from catalogue labels.
Product publication rejected with Only the owner can publish reviewed products.	Account lacks owner role	Save a draft and have the owner review/publish.

Message or symptom	Meaning	Correct next step
Publication requires reviewed English and Arabic content, source evidence and asset rights.	Required publication evidence is incomplete	Check both language names/descriptions, Source Ref, reviewer, date and Rights Confirmed.
Catalogue Details validation error	Invalid JSON shape/path/dimensions/bilingual values/date/link	Correct the relevant field against section 6. A remote URL belongs in sourceUrl, not image.src.
Product image still missing after valid save	Image path exists only as text, or wrong deployed file	Verify the actual deployed asset with the maintainer; the JSON validator does not upload files.
Published product does not show a new draft edit	Draft and public versions are separate	Owner must review and publish the changes; do not repeatedly create duplicates.
SKU identity is immutable	Existing definition was relabelled	Restore its original identity; create a new accurate SKU and deactivate the old one if needed.
Activate and review this exact SKU before receiving stock or creating a hold	SKU is inactive	Owner reviews Active state. Do not reactivate merely to bypass a legitimate retirement.
No eligible enquiry in hold selector	Not CMS, not genuinely verified, missing timestamp, closed, outside latest 100, or search mismatch	Check the private enquiry record and refine reference search. Do not fake verification.
SKU does not match selected enquiry line	Different product identity or direct model text mismatch	Review the line and exact SKU. The software does not infer substitutes.
Held and dispatched quantities would exceed this enquiry line	Requested quantity already allocated/shipped across one or more SKUs	Review all holds/dispatches for the line; release only genuinely unneeded remaining holds.
Insufficient available stock	Hold quantity exceeds On hand – Held – Blocked	Recheck balance/configuration; obtain a real receipt or revise the allocation.
Action would consume held or blocked stock	Adjustment would undercut committed units	Investigate count and commitments with owner/sales/warehouse. Do not delete reservations.
This action exceeds available or blocked stock	Too much blocked or unblocked	Check current Available and Blocked balances; enter the actual supported quantity.
Physical count does not match ledger	Full physical total differs from computed On hand	Investigate and document a justified adjustment before confirming.
Current count required before a hold	Freshness policy exists and confirmation is stale/missing	Owner or warehouse counts and confirms; reconciliation is not a substitute.

Message or symptom	Meaning	Correct next step
Freshness policy invalid	Environment setting outside its allowed format/range	Owner chooses policy; technical maintainer corrects configuration.
Reservation already released/dispatched/expired	No live remaining hold	Refresh and inspect its original, shipped, released and expired columns; do not backdate.
Quantity exceeds remaining held units	Partial action is larger than the remainder	Refresh and enter only actual supported units.
Ledger totals disagree/reconciliation failed	History is internally inconsistent	Stop stock writes, preserve evidence and ask owner/maintainer to investigate with backup protection.
Uncertain action or temporary inventory failure	Request may have committed despite missing response	Use Retry original action with the retained request key.
Enquiry Delivery Status remains not-configured	Legacy immutable field	Consult the matching Notification queue; current real sending remains disabled.
Download photo returns Not found	Unauthorized, invalid ID, expired/unreadable photo or unqualified enquiry	Confirm owner/sales session and metadata; ask maintainer if still within retention. The response intentionally does not disclose which private condition occurred.
Report has no rows	No demand matches selected dates/groups/visible labels	Reset filters; include test/demo only if those are intentionally being investigated.
Report request count differs from sum of row Requests	Same enquiry occurs in several groups	Use Enquiries total for distinct submissions.
CSV button missing	No rows or more than 2,000 groups	Narrow filters when oversized; do not interpret absent export as lost data.
Too many enquiry lines	Date period exceeds source limit	Shorten date period; a model filter alone cannot avoid this pre-filter cap.
Staff recovery unavailable	Activation prerequisites absent	Owner-assisted account support; do not promise email delivery or expose reset links.

14 How the stock implementation protects the business record

Every stock change is an append-only movement. A reservation preserves its original SKU snapshot, quantity, customer line, expiry, operator and reason. Partial releases/dispatches add movements rather than overwriting that original promise. The reservation's current state is calculated from these facts. That design supports later investigation of who received, held, shipped, freed or blocked units.

The service uses one PostgreSQL transaction/connection for its reads, locks and writes. A request-key lock prevents duplicate retry recording, an SKU lock prevents simultaneous stock changes from overselling the same balance, and a separate enquiry lock prevents different SKUs over-allocating the same requested line. A unique closure key permits one final reservation closure while allowing partial consumption records beforehand. Errors roll back the transaction.

All inventory collections deny ordinary create/update/delete access; hooks also reject trusted Payload local writes. Privileged raw database operators still have capabilities beyond these application restrictions. Investigation or repair must preserve a backup and the immutable business explanation; this is not authority to edit production history to make a number look right.

The private API is `GET/POST /api/staff/inventory`. Reads accept `skuId`, `skuSearch`, `enquirySearch`; writes require same-origin JSON, an authenticated staff session, an action-specific role, a unique UUID request key and the validated action fields. The browser manages that key automatically. A new accepted movement returns 201; an identical accepted retry returns 200. Old API release/dispatch callers without a quantity consume the full remaining hold; the current UI always asks for the explicit partial/full quantity.

This is a single aggregate balance per exact SKU. It is not a purchase-order, multi-warehouse, lot/serial, valuation, invoicing, payment or automated replenishment system. Those features cannot be inferred from a correct stock ledger.

15 Evidence limitations and source index

The following current discrepancies are documented rather than silently corrected in application code:

- Enquiries admin helper text still says reservations are inactive; inventory functionality now exists.
- Enquiry **Delivery Status** remains an immutable legacy `not-configured` value. Notification/verification queues are the implementation's delivery state.
- No catalogue Media collection or product-upload screen exists. Product media requires reviewed deployed files and Catalogue Details JSON.
- SKU definitions are readable by all staff, including catalogue editors; stock quantities and movement history are restricted to owner/sales/warehouse.
- Product unpublication is permitted to catalogue editors. Owner-only publication does not mean every public-content change is owner-only: categories have direct editable content too.
- Account deletion/full disable and last-owner protection are not implemented. Payload's separate default unlock access is broader than owner-only account editing.
- The inventory view/snapshot does not provide a separate manufacturer-part-number value, although SKU definitions store one.
- Broad historical paragraphs in the older catalogue-publication and progress documents describe earlier disabled functionality. Use the current milestone and checked source for today's scope.
- New/changed staff passwords have a stronger rule, while a pre-existing password may still be weaker. This task did not rotate it or inspect secret values.

No live staff mutation, product edit, SKU creation, enquiry, photo upload, stock movement, email, reset, scheduled job or deployment was performed for these chapters. The examples establish how the source behaves, not actual EL AMAL stock or customer demand. Exact commercial stock definitions, opening quantities, expiry agreements, freshness policy and final visual/business acceptance still require the owner's actual inputs. Monitoring remains deferred.

Source	Evidence supplied
src/payload.config.ts	Admin user collection, dashboard links, configured collections; absence of Media; PostgreSQL and email adapter integration
src/cms/collections.ts	Staff roles/auth, category/product/SKU fields, owner publication hook, access rules
src/lib/access.ts	Staff role helper and required product review evidence
src/lib/sku-identity.ts	Immutable SKU identity from creation
src/lib/catalogue-details.ts	Exact image/specification/availability/datasheet JSON validation
src/content/product-options.ts	Instrument/application option labels
src/lib/public-catalogue.ts	Explicit public projection, eligible categories/products and safe links
src/lib/load-catalogue.ts	Current published reads, request freshness and no demo fallback in CMS mode
src/app/(site)/[locale]/products/[slug]/page.tsx	Numeric CMS product identifier used as public route slug
src/cms/enquiries.ts	Immutable submitted facts, editable status/notes and stale helper/legacy delivery field
src/cms/notifications.ts, src/cms/verification-emails.ts	Queue labels, status values and owner/sales read-only access
src/cms/delivery-operations.ts	Owner-only maintenance collection
src/cms/verification.ts	Hidden server-only verification/request-limit records
src/cms/attachments.ts, src/components/admin/attachment-download.tsx	Enquiry photo metadata, protected download link and access
src/lib/attachment-service.ts, src/lib/enquiry-attachments.ts	Qualified photo storage/read, reconstruction, retention, quota and encryption
src/app/(frontend)/api/staff/enquiry-attachments/[id]/route.ts	Safe private download response and generic denial
src/components/inventory-console.tsx	Exact visible action/field labels, role action lists, pending retry and tables
src/components/inventory-admin-link.tsx, src/app/(staff)/staff/layout.tsx	Dashboard/staff navigation text and routes
src/app/(staff)/staff/inventory/page.tsx	Staff inventory entry gate

Source	Evidence supplied
<code>src/cms/inventory.ts</code>	Read-only reservation/movement fields and append-only hooks
<code>src/lib/inventory.ts</code>	Action permissions, request validation, fingerprint and quantity bounds
<code>src/lib/inventory-service.ts</code>	Transaction locks, balance/expiry/partial actions, freshness, eligibility and reads
<code>src/lib/inventory-http.ts</code>	Inventory API guards, statuses and retry-safe errors
<code>src/app/(staff)/staff/reports/page.tsx</code> , <code>src/components/demand-report-view.tsx</code>	Report entry gate and exact filter/table/export labels
<code>src/lib/demand-report.ts</code>	Query validation, cohorts, privacy labels, arithmetic and CSV escaping
<code>src/lib/demand-service.ts</code> , <code>src/lib/demand-http.ts</code>	Persisted role check, snapshot-only report query, bounds and private responses
<code>src/lib/staff-security.ts</code> , <code>src/lib/staff-recovery.ts</code> , <code>src/lib/staff-reset-lock.ts</code> , <code>src/lib/staff-email.ts</code>	Password validation, recovery gate, concurrency and provider adapter
<code>src/app/(payload)/api/[...slug]/route.ts</code>	CMS gate, blocked first registration and dedicated recovery routing
<code>node_modules/payload/dist/utilities/formatLabels.js</code> and <code>field/collection sanitizers</code>	Generated field/collection labels verified from installed dependency
<code>node_modules/@payloadcms/translations/dist/languages/en.js</code> and <code>UI SaveDraft/Select</code> components	English standard buttons, raw select-option labels and draft action semantics
<code>node_modules/payload/dist/auth/defaultUnlockAccess.js</code> , <code>auth/operations/unlock.js</code>	Separate default staff unlock permission
<code>PROGRESS.md</code> current milestone	Recorded live release, catalogue counts, zero exact SKU definitions, inactive intake/recovery and deferred monitoring
<code>docs/inventory-operations.md</code> , <code>docs/demand-reporting.md</code> , <code>docs/private-enquiry-photos.md</code>	Supporting runbooks, cross-checked with implementation
<code>docs/catalogue-publication-2026-09-28.md</code>	Historical catalogue evidence and folder-derived availability interpretation; historical activation paragraph is superseded

Backend security and operations handbook

This chapter describes EL AMAL as implemented at application commit

7f7326704305012e9637702b3112b5c85f554833, using the 29 September 2026 operational evidence recorded in PROGRESS.md. Documentation checkpoint f751ef5 follows that application release. It explains both the working application and the conditions still preventing a complete operational launch. Documentation preparation did not change application code, databases, accounts, deployments or credentials, and did not send email.

What the backend does

The backend is the part of the website that runs on the server. It reads the published catalogue, controls staff access, records enquiries, maintains stock history and prepares email. The browser presents those results, but cannot decide its own permissions, confirm its own email address, rewrite an enquiry or invent stock.

EL AMAL is one Next.js application with Payload CMS embedded in it. It is not a separate React frontend calling an independently deployed CMS. The public website, /admin, staff pages and HTTP API deploy together to the existing Vercel project. Payload provides the administration interface, authenticated staff accounts, collection permissions, database adapter and migration framework. Neon PostgreSQL holds the durable application records. The same source repository contains the interface, server services, migrations and verification scripts.

The application is an instrument catalogue and enquiry system. An enquiry is a request for technical or commercial review. It is not a paid order, quotation, invoice or automatic stock commitment. Email confirmation proves access to the enquiry's mailbox at confirmation time; it does not verify the customer's business identity or approve their requested configuration.

Current service inventory

Service or component	Purpose	Current state and boundary
Vercel	Hosts the Next.js application and daily cron	Existing project on the confirmed Hobby account; stable review site and administration are live.
Neon PostgreSQL	Stores catalogue, staff, enquiries, queues, inventory and private photos	Separate development and hosted databases; Neon is the installed Marketplace integration. Built-in Neon Auth is not the application's staff authentication.
Payload CMS	Staff login, collection administration, permissions and migrations	Embedded in the application; version 3.90.2.
Next.js / React / Node	Server-rendered website, routes and runtime	Source pins Next.js 16.3.5, React 19.2.8 and Node 22.x.

Service or component	Purpose	Current state and boundary
Resend adapter	Would deliver customer confirmations, staff notifications and password recovery	HTTPS implementation exists. No verified sender/domain and provider credentials are operationally configured; real customer and recovery delivery remain disabled.
Native Vercel cron	Bounded stock expiry and enabled retention	Registered daily at 02:00 UTC. It cannot provide the frequent email-worker health required for public intake.
Inngest	Proposed frequent scheduling option	Terms acceptance is outstanding; no resource was provisioned. An available free option is not an installed integration or an authorized scheduler.
Checkly / external monitoring	Proposed uptime and operational alerting	Explicitly deferred by Nour. No monitoring installation or external alert destination is claimed.
GitHub Actions	Checks source changes and disposable database behavior	Quality workflow runs on pushes and pull requests. It does not itself block Vercel's separate automatic deployment integration.
Local encrypted backup tooling	Captures and verifies a recoverable database snapshot	A local backup and isolated restore are verified. Automatic offsite backup scheduling and retention are not configured.

The receiving iCloud inbox is already confirmed. A receiving inbox is not a sending domain, proof of sender ownership, a provider API credential or a scheduler. Do not ask for the recipient again or treat the recipient setting as email activation.

How requests move through the system



Customer submission and email depend on activation. Reading the catalogue and authorized staff tools is already live.

The diagram shows implemented connections, including the currently gated mail path. It does not imply that an anonymous visitor can submit a live enquiry today.

Why this structure was chosen

Using one application keeps the public design, CMS records and staff workflows together without another deployment, API gateway or authentication service. PostgreSQL transactions make an enquiry and its queue entries succeed together, and allow stock checks to remain correct when several staff members act at once. Persistent database locks and leases coordinate different server instances; an in-memory JavaScript flag would protect only one instance and would disappear on restart.

The tradeoff is shared infrastructure. A database outage affects catalogue reads and staff operations. The configured database connection pool is deliberately small: at most three connections per Payload instance, with a 15-second connection timeout. Serverless replicas can still create more than three total connections. This is not evidence of unlimited capacity or a load-test result. External calls are kept outside long-held database transactions to avoid occupying scarce connections while waiting for an email provider.

The public catalogue uses an explicit projection: a new object containing public fields, rather than a raw CMS record. The trusted server read can inspect private publication evidence, but that evidence is not included in the website's public data. Products and categories paginate independently and concurrently; each collection's pages remain sequential. A failed read fails the request instead of silently returning a partial catalogue. Request-scoped caching deduplicates reads within a request while allowing later requests to see publishing and unpublishing changes.

CMS catalogue mode never falls back to convincing-looking demo products when the database is empty or unavailable. That protects the business from showing synthetic stock as genuine stock during an outage. Demo and real baskets also use separate storage keys.

Data ownership and access

Payload has 13 configured application collections. Physical PostgreSQL tables also include child arrays, versions, relationships, authentication and CMS support tables; the number of collections is not the number of tables in a backup.

Collection slug	What it represents	Who may read it through normal collection access	How it changes
staff	Login accounts and assigned role	Owner can read all; an authenticated non-owner can read their own record	Owner creates and updates; deletion denied. Password policy applies when a password is supplied.
categories	Bilingual catalogue groupings	All four staff roles	Owner and catalogue editor create/update; deletion denied. Public pages receive a separate projection.
products	Bilingual model records and reviewed technical details	Staff; anonymous collection access is restricted to published records	Owner and catalogue editor create/update, but only owner may publish; deletion denied. Draft versions enabled.

Collection slug	What it represents	Who may read it through normal collection access	How it changes
skus	Exact stock configurations	All four staff roles	Owner creates/updates; identity cannot be relabelled after creation; deletion denied.
enquiries	Immutable customer request snapshot plus staff workflow status	Owner and sales	Trusted submission service creates. Owner/sales can change business status and internal notes; submitted identity/content and confirmation fields are immutable through normal updates.
notifications	Staff notification queue	Owner and sales	Trusted server code only; direct create/update/delete denied.
enquiry-verifications	Confirmation digest, expiry and consumption evidence	No ordinary CMS role or anonymous reader	Trusted verification services only.
request-limits	Pseudonymous request/email allowance counters	No ordinary CMS role or anonymous reader	Trusted rate-limit services only.
verification-emails	Encrypted customer confirmation outbox	Owner/sales can read status metadata; encrypted envelope, delivery key and lease token explicitly denied	Trusted outbox and workers only.
delivery-operations	Scheduler lease, outcome, health and backlog counts	Owner only; lease token explicitly denied	Trusted scheduler only.
inventory-reservations	Immutable exact-SKU holds	Owner, sales and warehouse	Private inventory service; direct changes denied.
inventory-movements	Append-only stock event history	Owner, sales and warehouse	Private inventory service; direct changes denied.
enquiry-attachments	Private photo metadata and encrypted bytes	Owner/sales metadata only; sealedData and contentHash explicitly denied	Verified-customer upload service creates; retention service removes expired rows.

Public pages do not expose the internal SKU ledger. A catalogue label inherited from a supplied source page describes dated model or family availability, not an exact warehouse quantity. Production has no entered exact SKU definitions at the latest checkpoint. No zero-balance or opening-stock assumption should be converted into a customer promise.

Staff roles in practice

Task	Owner	Catalogue editor	Sales	Warehouse
Sign in to Payload administration	Yes	Yes	Yes	Yes
Read staff accounts	All	Own only	Own only	Own only
Create staff or change role	Yes	No	No	No
Edit category/product drafts	Yes	Yes	No	No
Publish reviewed products	Yes	No	No	No
Read SKU definitions	Yes	Yes	Yes	Yes
Create/change SKU definition	Yes	No	No	No
Use private inventory console and ledger	Yes	No	Yes	Yes
Receipt / adjustment / dispatch / block / unblock / physical count confirmation	Yes	No	No	Yes
Place or release a hold	Yes	No	Yes	No
Reconcile inventory	Yes	No	Yes	Yes
Read enquiries, update enquiry status/internal notes	Yes	No	Yes	No
Read demand reports and export CSV	Yes	No	Yes	No
Read private photo metadata/download	Yes	No	Yes	No
Read scheduler health	Yes	No	No	No

Inventory actions are checked in both the HTTP handler and the service. Buttons being hidden is not the permission boundary. The service validates the staff collection, role and numeric actor ID. A catalogue editor may see SKU definitions in the CMS while still being denied the inventory console and ledger.

Product publication requires English and Arabic names/descriptions, a source reference, reviewer, valid review date and confirmed rights. The publication hook rejects non-owner publication. A published product being edited by a catalogue editor must remain within the draft workflow; a normal update that attempts to leave the record published meets the same owner-only hook.

Staff authentication and recovery

New or changed passwords must contain 15-128 Unicode characters and be nonblank. A small explicit common-password list and repeated-single-character passwords are rejected. This is not a comprehensive breached-password database or proof that any chosen passphrase is strong. The previously chosen short owner password was not silently replaced; it still needs an authorized change to a strong unique password.

Payload is configured for five failed login attempts, a ten-minute lock and a two-hour token lifetime. Cookies use SameSite Lax and the Secure flag in production. Payload provides password hashing and session machinery; the application does not store recoverable staff passwords in ordinary collection fields. Public first-user registration is blocked both in the catch-all route and the collection endpoint. Initial owner creation is a trusted local bootstrap, not a visitor-accessible signup.

Recovery email is disabled until its explicit gate and sender configuration are complete. The custom recovery wrapper requires exact canonical HTTPS origin, JSON POST and a body no larger than 8 KiB. It applies four requests per minute to each visitor/operation scope and another one-per-minute bucket for a normalized staff email; Payload additionally sets a ten-minute minimum reset-request interval. The forgot-password response is neutral so it does not confirm whether an account exists. Reset tokens expire after 30 minutes, and a transaction-scoped advisory lock serializes competing uses of the same token through password update and session creation.

The staff email adapter sends directly through the HTTPS provider with a 15-second timeout and a body-derived idempotency key. It is not the durable customer outbox. Its existence does not make password recovery operational. There is no implemented application MFA enrollment flow and no completed penetration-test certification. MFA in hosting/provider accounts is a separate control and cannot be inferred from staff login settings.

HTTP route reference

These are the concrete application route contracts at the current source revision. Authentication failure, disabled configuration and invalid input can intentionally return generic responses. API and staff paths receive private/no-store and noindex headers. A 200 page containing a sign-in explanation is not evidence that protected data was returned.

Application-owned API routes

Method and path	Intended caller and input	Result and important conditions	Implementation
GET /api/customer-enquiries	Browser checking configured intake availability	{canSubmit}. This is configuration/origin presence only; POST additionally checks real durable worker health.	src/lib/customer-http.ts, customer-readiness.ts
POST /api/customer-enquiries	Anonymous customer, once enabled; exact canonical origin; JSON enquiry	202 with random reference, signed resend receipt, awaiting-verification status, emailSent:false, stockReserved:false. 400 input, 403 origin, 409 changed request-key content, 415 type, 429 rate, 503 unavailable/health/capacity/error.	customer-http.ts, customer-service.ts, submit-enquiry.ts

Method and path	Intended caller and input	Result and important conditions	Implementation
POST /api/customer-enquiries/resend	Customer holding signed receipt; exact origin; JSON receipt, maximum 4 KiB	Neutral 202 accepted for invalid/expired/ineligible receipts and email-limit/capacity outcomes; visitor limit can return 429. Never accepts a replacement email address.	customer-http.ts, customer-service.ts
GET /api/enquiry-submissions	Staff test UI	{canSaveTest} only when CMS is configured, catalogue source is demo and caller is owner/sales.	enquiry-http.ts
POST /api/enquiry-submissions	Owner/sales testing demo enquiry; same origin, JSON	201 new or 200 repeated test save; no mail or stock reservation. Unavailable in real CMS-catalogue mode.	enquiry-http.ts
POST /api/enquiry-verification	Holder of email token; same origin; JSON token, maximum 4 KiB	Single-use atomic confirmation. Returns reference, customer/test mode and stockReserved:false; customer mode may include upload grant when enabled. 400 invalid/expired/used link; 429 rate; 503 unavailable.	verification-http.ts, enquiry-verification.ts
POST /api/enquiry-verification/test-link	Authenticated owner/sales; demo enquiry reference	Returns a test token only for an eligible unverified demo enquiry. Never issues a public customer token.	verification-http.ts, verification-runtime.ts
POST /api/customer-photos	Verified customer holding bearer upload grant; exact origin; raw JPEG/PNG body	Signed capability plus database confirmation checks. Uses X-Upload-Id and encoded X-Photo-Name; not multipart. 403 unauthorized, 415 type, 429 rate, 413 oversized stream, 400 invalid/content/quota failure, 503 disabled.	attachment-http.ts, attachment-service.ts
GET /api/staff/enquiry-attachments/:id	Authenticated owner/sales; positive numeric ID	Private attachment download. Generic 404 for invalid ID, denied caller, expired/unreadable file or unavailable CMS.	Staff attachment route.ts, attachment-service.ts

Method and path	Intended caller and input	Result and important conditions	Implementation
GET /api/staff/inventory	Owner/sales/warehouse	Private inventory view; 403 unauthorized, 503 unavailable.	inventory-http.ts, inventory-service.ts
POST /api/staff/inventory	Authorized role for the command; same-origin JSON, maximum 8 KiB	Validates command/role and executes transactionally. 201 new action, 200 exact repeated action, appropriate input/conflict denial.	inventory-http.ts, inventory.ts, inventory-service.ts
GET /api/staff/demand-report	Owner/sales; validated date/filter/query options	JSON report or downloadable CSV. Private /no-store/no-referrer. Contact details and arbitrary customer text excluded.	demand-http.ts, demand-report.ts, demand-service.ts
POST /api/internal/notifications	Authorized machine bearer using NOTIFICATION_WORKER_SECRET	At most one eligible staff notification. 401 before database access if unauthorized; 503 disabled/error.	notification-http.ts, notification-worker.ts
POST /api/internal/verification-emails	Same machine secret; verification delivery enabled	At most one customer confirmation job; 401 before database access or 503 disabled/error.	notification-http.ts, verification-email-worker.ts
GET or POST /api/internal/delivery-operations	Authorized scheduler bearer using separate CRON_SECRET	Bounded batch and maintenance. 200 complete/maintenance/overlap; 503 degraded/disabled/stale/error; 401 bad authorization. This performs work and is not a public health probe.	delivery-operations.ts, delivery-runner.ts
HEAD /api/internal/delivery-operations	Any caller	Explicit 405. Prevents Next.js automatically running GET work for HEAD.	Operations route.ts
GET /social-image	Public social-image fetch	Generated social artwork; outside the private /api namespace.	src/app/social-image/route.tsx

Each API route file is under `src/app/(frontend)/api/` at its corresponding path. Unsupported methods are not an alternative way to invoke these services. The one exception to POST-only mutation is the explicitly bearer-protected operations GET required by native cron.

Payload REST surface

`src/app/(payload)/api/[...slug]/route.ts` dispatches GET, POST, PATCH, PUT, DELETE and OPTIONS to Payload after the CMS configuration gate. Dispatching a verb does not mean every collection permits that operation. Collection/field access rules above still apply. GraphQL is explicitly disabled.

For each listed collection slug, Payload's installed REST endpoint definitions include:

Path pattern	Methods and meaning
<code>/api/:collection</code>	GET find; POST create; PATCH bulk update; DELETE bulk delete, subject to collection permissions.
<code>/api/:collection/:id</code>	GET record; PATCH update; DELETE delete, subject to permissions.
<code>/api/:collection/count</code>	GET count subject to read access.
<code>/api/:collection/access/:id?</code>	POST document-access calculation.
<code>/api/:collection/:id/duplicate</code>	POST duplicate endpoint, subject to collection create/access rules.
<code>/api/:collection/versions</code> and <code>/versions/:id</code>	GET version APIs, meaningful for version-enabled products; POST <code>/versions/:id</code> restores a version under framework permissions.
<code>/api/access</code>	GET framework access information.

The authenticated `staff` collection also registers GET `/api/staff/init`, GET `/api/staff/me`, and POST `/api/staff/login`, `/logout`, `/refresh-token`, `/forgot-password`, `/reset-password`, `/first-register`, `/unlock`, and `/verify/:id`. These are framework routes with framework preconditions, not a promise that every optional feature is configured. Staff email verification and API-key authentication are not enabled in the collection configuration. First-register is explicitly denied; forgot/reset are intercepted by the custom guarded recovery wrapper. The principal staff entry point is `/admin/login`, not these raw API endpoints.

The installed endpoint definitions can be inspected in `node_modules/payload/dist/auth/endpoints/index.js` and `node_modules/payload/dist/collections/endpoints/index.js`. They are dependency source, not files maintained by this project. Do not add undocumented authentication behavior based on a different Payload version.

Customer enquiry lifecycle

What a customer would do after activation

1. Browse the real English or Arabic catalogue and add products/quantities to the quote basket, or enter a direct model, quantity and required range.
2. Enter name, company, email and optional technical notes. Review the request before submitting.
3. Receive a random enquiry reference once the server atomically saves the request and its required queue records. The immediate response explicitly says email has not yet been sent and stock has not been reserved.
4. Receive a confirmation email, open the link and actively press Confirm. Merely opening the link does not consume it.

5. After genuine confirmation, optionally upload eligible technical photos on that page. Staff become eligible to receive the reference-only notification.
6. Sales review the private enquiry and contact the customer through an approved business process. A staff member decides any configuration, quote and exact-SKU stock action.

That customer journey is implemented but public intake is still gated off. The website can still be browsed and requests reviewed without claiming a working submission channel.

Submitted data and validation

An enquiry contains a random version-4 UUID request key, English/Arabic locale, contact object and either catalogue basket lines or a direct-model request. The contact fields are name up to 120 characters, company up to 160, email up to 254 and notes up to 2,000. Name/company/email are required; email is normalized to lowercase for customer intake and checked server-side.

Catalogue input allows at most 100 unique product IDs, each up to 100 characters, with whole-number quantities from 1 to 9,999. The server resolves those IDs against its current public catalogue and snapshots model plus English/Arabic names. It does not trust caller-provided product descriptions or statuses. Direct RFQ input requires a model up to 120 characters, nonempty range up to 160 and the same quantity bounds; it cannot also include catalogue lines. A direct request is labelled customer-specified and needs technical review, not treated as an existing SKU.

New submissions are streamed with a 32 KiB body limit, checked independently of a claimed content length. They require JSON and an Origin header matching the configured canonical HTTPS origin and actual request origin. This makes customer intake intentionally unavailable from arbitrary mirrored or protected-preview origins unless configured deliberately for that environment.

Duplicate protection and durability

The browser creates a request key for a reviewed submission. The server hashes the normalized request content and stores both that fingerprint and the unique key. Retrying the identical key/content returns the original reference. Reusing the key with different content returns a conflict so a changed request cannot silently replace the first request.

The enquiry snapshot, one staff-notification record and the customer verification digest/encrypted outbox are created in the same database transaction. A queue failure rolls back the enquiry instead of leaving a customer believing a deliverable request exists. Database uniqueness also handles simultaneous identical submissions. A successful saved retry is resolved before re-checking changed catalogue data, queue capacity or worker freshness, so an outage does not hide an already accepted request.

This protects against accidental double submission, but does not give every business event universal exactly-once delivery. A third-party provider can accept a message while a network timeout prevents the application recording that fact. Stable provider idempotency keys and bounded retry windows address that separate uncertainty.

Confirmation and resend are different capabilities

The customer verification token is a cryptographically random 32-byte value represented as 64 hex characters. The database verification record stores only its SHA-256 digest. It expires after one hour and is consumed atomically with confirmation. Demo confirmation produces `test-verified`; only genuine CMS confirmation produces `verified` and can qualify for customer photos or staff notification delivery.

The emailed link carries its token in the URL fragment. Fragments are not sent in ordinary HTTP requests. The page reads it into memory, requires the user to press Confirm and removes the fragment before the confirmation API request. This reduces accidental consumption by link-preview fetchers; it does not turn an email token into a public reusable link.

The immediate submission receipt is a separate HMAC-signed, resend-only capability lasting 24 hours. It cannot confirm an address, read the enquiry or select another recipient. The current UI stores it only in memory; refresh or tab closure loses resend controls. Resending replaces the old digest/message, so the earlier confirmation link stops working. Requests within the one-minute cooldown or after three confirmation generations in a 24-hour window are not issued another link. A message already in flight cannot be recalled; an older email may arrive with a link that is now invalid.

Abuse limits and queue admission

- Customer submission and resend use six requests per minute per pseudonymous visitor/scope.
- A normalized email has six new submission/resend allowances per fixed 24-hour window; this counts eligible attempts, not guaranteed emails sent. Repeating the same attempt key does not consume another allowance.
- Request/email keys use purpose-separated HMACs; raw IP addresses and email addresses are not stored in the rate-limit table.
- The forwarded IP is trusted only when the application is running directly behind the configured Vercel proxy. Other hosts or invalid headers use a shared fallback bucket.
- Queue admission takes a transaction-scoped advisory lock and refuses new CMS work when either active confirmation jobs or active CMS staff jobs has 100 rows. This bounds active backlog, not lifetime record count or all platform costs.
- New requests require the durable delivery row to show outcome complete and a nonfuture success timestamp no more than 15 minutes old. A missing/stale/degraded heartbeat refuses new persistence before the email allowance is consumed.

These limits make ordinary abuse more expensive and bound some resource use. They do not demonstrate resistance to all distributed abuse, DDoS, malicious staff activity or future vulnerabilities. There is no claim of a CAPTCHA, independent WAF configuration or penetration test in this application release.

Email queues scheduling and operational health

Two queues with different jobs

The customer outbox contains a snapshot of sender, recipient, bilingual message and confirmation link. AES-256-GCM encrypts that envelope with a purpose-specific HKDF-derived key from `PAYLOAD_SECRET`. A fresh nonce is used, and authenticated associated data binds ciphertext to its delivery generation. Sent or terminal records erase the replayable envelope while preserving delivery metadata.

The staff queue sends only the enquiry reference and an instruction to sign in. Customer notes, contact details and attachments are not copied into the notification body. Staff delivery waits until the related CMS enquiry has genuine verified status and a confirmation timestamp. Waiting for confirmation does not consume attempts; demo records are permanently disabled. Customer enquiry data still exists in the authorized private inbox independently of an email's transport outcome.

Workers claim records atomically using database row locks with `SKIP LOCKED`, increment attempts, and assign a random five-minute lease. Other workers can continue with different rows. Expired claims can be recovered, and late results must match the current lease before changing the row. Provider calls use HTTPS, reject redirects, abort after 15 seconds, supply a stable idempotency key and accept only a bounded valid provider receipt. Provider error bodies are not retained in user-facing errors or queue diagnostics.

Retries use exponential delays beginning at 60 seconds, with a five-attempt ceiling. Staff notifications are not retried beyond 23 hours from creation, preserving the conservative window chosen for provider deduplication. Customer confirmation claims require more than 30 seconds of token life remaining. A provider acceptance receipt is not proof of inbox delivery or customer confirmation.

The batch runner

The protected operations route has a separate two-minute scheduler lease so overlapping calls return `overlap`. One invocation starts at most eight message attempts, alternates queues and stops starting more work after a 40-second budget that begins before CMS initialization. The route's duration limit is 60 seconds. Leases and idempotency handle repeats and interrupted invocations; successful HTTP scheduling alone is not enough.

Maintenance can operate with sending disabled. It expires at most 25 holds with a five-second start budget, cleans bounded secret/rate-limit data when retention is enabled, and removes at most 50 expired photos. A maintenance-only outcome has zero email attempts and never advances the successful-mail timestamp. Expired holds already stop reducing availability even when physical cleanup waits for the daily invocation.

The owner-only delivery operations collection shows last start/completion/success, last outcome, attempt/failure counts, queue sizes and oldest pending work. This is durable internal health, not an external monitoring service. A healthy empty-mail-queue pass can establish worker activity; a daily maintenance-only pass cannot satisfy the mail gate.

Why the existing daily cron cannot activate customer intake

The current Vercel schedule is 02:00 UTC once per day. Its purpose is bounded housekeeping on the confirmed Hobby project. New customer enquiries require successful mail-worker evidence within 15 minutes and confirmation links expire in one hour. A daily schedule cannot maintain either practical delivery cadence or continuous fresh health. A separately authorized frequent scheduler, normally every five minutes, remains required; upgrading a plan or accepting integration terms was not done implicitly.

Do not use the local laptop as permanent production scheduling. Do not count GitHub scheduled workflows as a guaranteed timely email service. Inngest terms have not been accepted, and external Checkly monitoring was explicitly deferred. Those facts must stay visible instead of being converted into a completed checklist item.

Activation sequence once prerequisites are supplied

1. Confirm the owned sending domain/address, actual provider account and canonical HTTPS origin. Store credentials privately. The receiving inbox is already known.
2. Confirm migration state and runtime access in the intended environment. Inspect readiness booleans and the private queue without exposing customer data.

3. Configure and authorize the frequent scheduler and protected bearer secret. Enable worker transport and verify two healthy scheduled outcomes with fresh timestamps; daily maintenance is insufficient.
4. Prepare an authorized controlled customer test, including permission to send real email. Intake must be enabled in the controlled environment/window to exercise its actual API; preserve the launch gates until this test is ready.
5. Verify actual receipt of the customer email, successful one-time confirmation, staff notification receipt in the confirmed inbox, and expected private records. Test resend/old-link invalidation and failure/retry behavior with appropriate test accounts.
6. Set the manual workers-ready assertion only after the real scheduler/delivery evidence exists. Open public intake after controlled acceptance and continue watching the durable health row.
7. Enable and verify staff recovery separately. Its direct adapter is not proved by a customer queue test.

`npm run readiness` makes no network calls and prints configuration booleans only. It cannot prove domain ownership, provider account access, real receipt, scheduler activity, stock correctness or client launch approval.

Operator response to delivery trouble

Investigate any operations 503, successful-mail timestamp older than 15 minutes, increasing backlog, customer queue older than ten minutes or staff work approaching 23 hours. First distinguish disabled configuration from maintenance-only health, provider failure and an interrupted invocation. Compare timestamps and lease expiry; an old displayed result can survive a timed-out worker.

During a prolonged outage, pause new public intake while allowing authorized workers to recover pending work. Preserve records. Check provider receipts before deciding whether a timed-out message was accepted. Do not reset attempts, rewrite creation dates or reuse an old job beyond its deduplication window. Do not change sender, recipient or template while retryable staff jobs are pending without reconciling the queue. Turning off public intake alone does not drain or erase existing queues.

Private technical photos

Photos are an implemented, deliberately narrow attachment workflow. They are not general document upload. A genuinely verified CMS customer can upload at most three JPEG/PNG still images, each up to 2 MiB (2,097,152 bytes). PDF, XLSX, SVG, video and arbitrary documents are not enabled; neither a 10 MB limit nor malware-scanned document quarantine is implemented.

The confirmation response can issue a signed enquiry-specific upload grant lasting 24 hours. The browser holds it in memory and sends it as a bearer header. It is not a public download URL, and the application has no later customer attachment portal. Refreshing the confirmation page loses the grant; the consumed original email token cannot simply be reused to obtain it again.

The upload route streams and bounds actual bytes, verifies declared MIME against signature, and decodes the image with an 8,000,000-pixel limit. It applies orientation, resizes inside 2400 by 2400 without enlargement and writes a fresh JPEG or PNG that must still fit the byte cap. Rebuilding the pixels discards embedded metadata and appended content. It does not scan arbitrary documents or remove private information visibly photographed in the image.

Reconstructed bytes are encrypted with AES-256-GCM and a fresh nonce, using a purpose-separated key derived from `PAYLOAD_SECRET`. Authentication binds a photo to its enquiry reference and upload UUID. A transaction and advisory lock serialize per-enquiry counting and the shared quota: 64 MiB of

reconstructed bytes across the table. Base64 encryption envelopes and database/index overhead mean physical database size can exceed that logical cap. The quota includes expired rows until cleanup deletes them.

A repeated UUID with the same enquiry, reconstructed content hash, sanitized filename and MIME returns the previous save. Changing the file while reusing the UUID is rejected. The browser preserves the UUID during a retry of the same selected file. A failed response should therefore be retried with the same file and identity, not by generating many new attempts.

Only owner/sales may download. The download requires staff authentication and current genuine enquiry confirmation; it returns a generated filename, attachment disposition, no-sniff, private/no-store and restrictive default-src 'none'; sandbox policy. Invalid, unauthorized, expired and unreadable cases share a generic not-found response. Encrypted contents never enter public/ and are not emailed as attachments.

Photos become unavailable exactly 30 days after creation. Physical deletion requires enabled retention and a successful operations invocation. Backups can retain older encrypted copies; the live 30-day policy is not automatic erasure from every backup. When the quota fills, new admission stops instead of deleting active photos or silently expanding storage.

The current encrypted format has no key ID or dual-key reader. Changing PAYLOAD_SECRET invalidates outstanding grants and makes old photo/outbox ciphertext unreadable with the new secret. A planned rotation needs paused dependent workflows, an encrypted backup, preserved old secret in private custody, and either a reviewed re-encryption migration or explicit expiry/retention plan. No automated re-encryption tool is included. Losing the old secret cannot be repaired by restoring only the database.

Security controls and their limits

Implemented controls

Risk	Relevant implemented control	Practical limit
Unauthorized staff/data access	Payload role/field access plus service authorization; first-register denied; private routes	Staff roles remain trusted within their scope; compromised credentials can use their permissions.
Cross-origin mutation	Exact/same-origin checks on custom mutation handlers, JSON requirements, production secure cookies	Do not describe this as a custom universal CSRF token system; Payload's ordinary routes use framework behavior.
Input abuse	Bounded streaming bodies, UUID/token patterns, quantities, names and command parsing	Not every framework route has the same custom body/rate guard.
Duplicate or concurrent writes	Unique request keys, fingerprints, transactions, row/advisory locks and leases	Third-party outcomes remain potentially ambiguous; provider reconciliation matters.
Email/token disclosure	Digest-only confirmation records, encrypted outbox, private responses, reference-only staff mail	Authorized enquiry contact fields are not application-encrypted; application compromise with secrets defeats envelope protection.

Risk	Relevant implemented control	Practical limit
Upload abuse	Strict JPEG/PNG limits, full decode/re-encode, signed grants, verified enquiry, quota and private download	Not antivirus scanning, content moderation or support for arbitrary documents.
Query injection	Parameterized runtime queries and validated identifiers; collection permissions	A source review and tests are not a comprehensive penetration test.
Dependency advisories	Lockfile, pinned package versions, narrow compatibility overrides and all-severity CI audit	Zero known findings at a checkpoint is not absence of undiscovered vulnerabilities.
Accidental indexing/caching of private data	Private/no-store, no-referrer and noindex on administration/staff/API; robot exclusions	Robots/noindex do not authenticate or prevent a determined visitor requesting a URL.

Global headers disable MIME sniffing and framing, constrain objects/base URI/forms, restrict camera/microphone/geolocation and add HSTS for one year. The global CSP is `object-src 'none'; base-uri 'self'; frame-ancestors 'none'; form-action 'self'`. It has no restrictive `script-src` or per-response nonce policy and should be described as baseline hardening, not comprehensive XSS prevention. Public referrers use `strict-origin-when-cross-origin`; private routes are overridden to `no-referrer`. HSTS is not configured with `includeSubDomains/preload` in this source. The `powered-by` header is disabled.

Known source limitations to address after documentation

- Notification worker metadata is only hidden in the admin UI.** `src/cms/notifications.ts` marks `deliveryKey` and `leaseToken` with `admin.hidden` but does not deny their field-level read access. Owner/sales can therefore potentially receive those values through their authorized collection reads. Anonymous readers are denied the entire collection. These values are an idempotency identity and an internal lease identity; neither authenticates a public worker endpoint. The worker routes separately require `NOTIFICATION_WORKER_SECRET` or `CRON_SECRET`, and direct queue writes are denied. No anonymous exposure or direct worker takeover is established. The minimal future change is field read denial plus owner/sales denied-field assertions and a trusted-worker regression.
- Enquiry admin helper text is stale.** The collection description still says stock reservation is not active, although private staff stock operations exist. Its immutable `deliveryStatus` field is still a legacy not-configured value; actual email state lives in the queue collections. Operators should consult those queues, not use that field as delivery telemetry.
- No application MFA or penetration-test acceptance is implemented.** Strong-password policy, login lockout and recovery controls are real safeguards but do not establish either claim. Existing owner credentials still need strengthening.
- No independently provisioned external alerting.** Owner-visible health exists; Checkly remains deferred. A failed daily job can require manual discovery until an approved external alerting arrangement exists.
- No automatic offsite recovery schedule.** The successful local archive/rehearsal establishes one recoverable snapshot, not a continuous recovery service.

6. **The broader upload requirement remains open.** Larger PDF/Excel private quarantine, scanning and storage need their own design, implementation and verification.
7. **Staff unlock inherits a broader framework default.** The staff collection does not define `access.unlock`. Installed Payload 3.90.2 fills this with `defaultUnlockAccess`, which allows an authenticated member of the configured admin collection. Because all four staff roles belong to that collection, source inspection indicates that a signed-in non-owner can use the framework unlock operation for another staff email. Anonymous access remains denied; unlock clears login-attempt/lock state and does not reveal or change a password or grant a role. This is broader than an owner-only account-administration policy. The minimal future correction is explicit owner-only unlock access, verified with owner/non-owner/anonymous requests against disposable accounts and with normal login lockout preserved. This finding is from source inspection, not a production mutation or a newly executed exploit test.

No application fix was made while producing this chapter. These are concrete follow-up findings, not a claim that undocumented hardening already happened.

Configuration reference

Values, passwords, tokens, private hostnames, provider credentials and connection strings must stay outside this handbook. The table lists variable names and purposes only. `.env.example` is a starting reference, not proof of hosted configuration. Vercel may intentionally return sensitive values blank during an environment pull; a blank downloaded value alone does not prove the deployed setting is missing.

Variable name	Purpose and dependency
DATABASE_URL	Application database connection; development and hosted targets must remain distinct.
DATABASE_URL_UNPOOLED	Direct database connection for workflows that cannot use a transaction pooler.
PAYLOAD_SECRET	Payload authentication plus signed/encrypted application capabilities; minimum length gate and coordinated rotation required.
CMS_ENABLED	Enables configured CMS runtime; also requires database and secret.
CATALOGUE_SOURCE	Selects reviewed CMS catalogue versus explicit demo mode.
SITE_URL	Canonical site origin; customer/recovery mail requires exact HTTPS origin without path, credentials or trailing slash.
PUBLIC_ENQUIRIES_ENABLED	Public customer submission gate, currently disabled.
ENQUIRY_WORKERS_READY	Operator assertion after actual scheduler/delivery verification; does not replace durable health.
ENQUIRY_NOTIFICATION_TO	Confirmed staff receiving destination; not a sender credential.
NOTIFICATION_DELIVERY_ENABLED	Enables mail transport readiness for staff/customer workers.
VERIFICATION_DELIVERY_ENABLED	Additional customer confirmation delivery gate.
MAIL_PROVIDER	Selects the implemented provider adapter.

Variable name	Purpose and dependency
MAIL_FROM	Verified-domain plain sender address; requires actual provider/domain verification.
RESEND_API_KEY	Provider authorization credential; keep private.
NOTIFICATION_WORKER_SECRET	Separate machine bearer for the one-item worker endpoints; not a staff password.
DELIVERY_OPERATIONS_ENABLED	Enables the scheduled batch/maintenance endpoint.
DELIVERY_RETENTION_ENABLED	Separately enables expired secret/rate/photo cleanup.
CRON_SECRET	Independent protected scheduler bearer.
STAFF_RECOVERY_ENABLED	Staff password-email recovery gate, currently disabled.
ENQUIRY_PHOTOS_ENABLED	Enables photo grant/upload capability; genuine confirmation remains required.
INVENTORY_FRESHNESS_HOURS	Optional whole-hour freshness policy; absent setting leaves review manual.
SITE_INDEXING_ENABLED	Production real-catalogue indexing gate; not a ranking guarantee.
GOOGLE_SITE_VERIFICATION	Optional Search Console ownership token.
BUSINESS_PHONE, BUSINESS_WHATSAPP	Approved public contact information.
WIKA_RELATIONSHIP_EN, WIKA_RELATIONSHIP_AR, WIKA_EVIDENCE_URL	Approved relationship wording and documentary evidence.
BACKUP_DATABASE_URL	Optional explicit direct source for encrypted backup.
PG_BIN_DIR	PostgreSQL command-line tool location.
BACKUP_KEY_FILE, BACKUP_ENCRYPTION_KEY	Private separate backup-key source; file-based local custody or protected automation secret.
RESTORE_DATABASE_URL, RESTORE_ALLOWED_HOST	Explicit separate development destination authorization for isolated restore rehearsal.
CMS_DATABASE_CHECK	Development-only regression guard; not a bypass permitting production test writes.

Platform/runtime inputs such as `VERCEL`, `VERCEL_ENV`, `NODE_ENV`, `CI` and `NEXT_TELEMETRY_DISABLED` also affect proxy trust, indexing, cookies or checks. They are not user credentials. Maintain separate environment scopes and verify runtime behavior after changes; do not bulk-copy development configuration into production.

Database migrations backups and recovery

Schema changes

Payload's Postgres adapter has automatic schema push disabled. Changes are recorded as versioned migrations under `src/migrations`, registered in `src/migrations/index.ts`. The ten current migrations cover the initial catalogue, security upgrade, enquiry inbox, staff queue, client requirements, confirmation, encrypted confirmation outbox, detailed catalogue, launch operations and inventory completion.

The latest is `20260929_141342_inventory_completion`; the preceding operations migration is `20260929_002849_launch_operations`. Both were applied to development and hosted databases according to the current progress evidence. A code deployment and a database migration are separate events. A Ready deployment is not proof that its required schema exists. Down migrations may drop data and are not the normal response to an application rollback.

The safe sequence is review the migration, rehearse on development/disposable data, take and verify an appropriate backup, apply the reviewed forward migration to the intended target, then verify compatible application behavior. Additive migrations reduce rollback friction but do not make every older version compatible automatically. Never test integration scripts against hosted production to avoid setting up a development environment.

What the verified backup proves

The encrypted local archive is `artifacts/backups/el-amal-2026-09-29-verified.enc`, accompanied by `.manifest.enc`. The restore report records **2026-09-29T14:29:27.653Z, 24 public tables, 572 rows**, matching sorted row fingerprints and matching constraint count; the random temporary development database was removed. The backup predates the latest inventory-completion migration. It must be restored with the matching source/schema context and then advanced with reviewed forward migrations where required.

The backup script shares a repeatable-read exported snapshot between its manifest and PostgreSQL custom-format dump, so the two describe one database point in time. AES-256-GCM protects archive and manifest with independent nonces, and the manifest binds the archive SHA-256. Rehearsal authenticates the full archive before `pg_restore`, restores into a newly named database on the explicitly allowed different unpooled development host, and compares rows plus constraints. Plaintext is temporarily created under a random operating-system temporary directory and cleanup attempts all owned resources even after failure.

The backup key is held separately outside the OneDrive project. Neither its value nor other application secrets belong in the archive documentation, chat or Git. Keep both encrypted files and the appropriate original `PAYLOAD_SECRET` in separate approved custody: encrypted photo/outbox contents require that Payload secret as well as the database backup key.

The archive covers the public application schema. It does not back up Neon account settings, cluster roles, DNS, provider accounts, local secrets or every hosting configuration. Bundled product media is recovered from the matching Git source revision. GitHub preserves committed source, not live database data. OneDrive placement of an encrypted local artifact is not a configured, verified offsite backup policy with independent retention and alerts.

Documented backup commands

These commands are an operator reference, not commands executed during handbook preparation. Their named ignored environment files must already contain the correct private configuration. Use PostgreSQL tools matching or newer than the hosted server major version; the recorded hosted database uses PostgreSQL 18.

```
node --env-file=.env.hosted.local --env-file=.env.backup.local scripts/database-backup.mjs create
artifacts/backups/el-amal-YYYY-MM-DD.enc
node --env-file=.env.backup.local scripts/database-backup.mjs rehearse
artifacts/backups/el-amal-YYYY-MM-DD.enc
```

Choose a new filename; the script refuses overwrite. Preserve both encrypted outputs and the unique restore report. A successful archive command without a successful rehearsal is not equivalent to the verified snapshot above. Scheduling, offsite destination, retention period, acceptable data-loss window and recovery-time target still need explicit operational setup and acceptance.

Actual incident recovery

1. Preserve the affected database and its logs for diagnosis. Identify the incident time, latest recoverable archive and matching application revision.
2. Retrieve encrypted archive and manifest plus the separately held backup key and required original application secret through the approved private process.
3. Restore to a separate replacement database under an operator's supervision; never point rehearsal at production or erase the affected database in place.
4. Verify table contents, constraints, staff access, published catalogue, private records and encrypted downloads. Apply reviewed forward migrations after establishing the correct restored baseline.
5. Point a protected candidate deployment at the replacement and perform runtime checks before switching production.
6. Reconcile any business/email events after the backup timestamp with provider or staff records. A database restore can roll back recorded delivery outcomes while an email has already been sent.
7. Switch only after recovery acceptance, observe the restored service, document the real data-loss/time outcome, and preserve incident evidence.

No complete automated production failover or point-in-time restore guarantee has been established. The local rehearsal is strong evidence for that archive, not an unmeasured promise about recovery time.

Development CI and deployment operations

Working locally

Use the existing project and Node version from `package.json`. `npm ci` installs the lockfile, `npm run dev` starts the loopback development server on port 3004, `npm run build` makes a production build and `npm start` serves it on the same loopback port. Clean builds can download configured fonts. Local development must use the development database; production credentials are not a convenient shortcut.

The installed Next.js version requires its bundled documentation for coding changes. `AGENTS.md` points to `node_modules/next/dist/docs/`; use that source rather than relying on assumptions from earlier Next.js releases. The current application deliberately uses Webpack for dev/build. Payload's admin import map

uses a resolved filesystem path because a previously bundled URL form built successfully but failed at runtime.

Windows host memory has interrupted earlier parallel checks and local builds. Run resource-heavy build/database tasks serially, stop unneeded owned processes, and record which environment actually completed a check. A terminated local build is not a successful build; the verified Linux CI/Vercel builds provide the release build evidence. Do not disable assertions simply to fit the host.

Verification commands and boundaries

Command or script	What it verifies	Operational caution
<code>npm test</code>	Unit/permission/validation/crypto/hel per regressions	142 checks succeeded at the current application checkpoint; new runs may change the count.
<code>npm run typecheck</code>	Integrated TypeScript consistency	Type success does not prove browser or database behavior.
<code>npm audit --audit-level=low</code>	Known dependency advisories at all severities	Recorded result is zero; must be rechecked for future releases.
<code>npm run readiness</code>	Presence/shape of configuration gates	No network, no secrets and no real readiness/receipt proof.
<code>npm run build</code>	Production compile/build	Needs runtime smoke checks afterward.
<code>scripts/check-cms-database.ts</code>	CMS permissions, draft/publication and projection behavior	Development guard required; use private development settings.
<code>scripts/check-enquiries.ts</code>	Request persistence/deduplication/queue transaction behavior	Development database and disposable fixture cleanup.
<code>scripts/check-verification.ts</code>	Confirmation behavior	Synthetic fixture data, no real customer message.
<code>scripts/check-notifications.ts</code>	Staff queue claims/retries/leases/eligibility	Fake transport, no provider contact.
<code>scripts/check-verification-emails.ts</code>	Encrypted outbox, resend, token rotation and worker behavior	Fake transport; not actual inbox evidence.
<code>scripts/check-customer-intake.ts</code>	Quotas, health gating, retries/conflicts and request persistence	Isolated development schema; does not overwrite real health to make a test pass.
<code>scripts/check-request-limit-cleanup.ts</code>	Cleanup versus concurrent quota-renewal regression	Development-only concurrency test.
<code>scripts/check-delivery-operations.ts</code>	Batch leases, queue capacity, retention and genuine verification gate	Disposable schema; fake mail transport.
<code>scripts/check-staff-and-photos.ts</code>	Photo crypto/quota/access/retention and real Payload reset/session behavior	Checks intended development/hosted separation, uses owned isolated schema and fake mail.

Command or script	What it verifies	Operational caution
<code>scripts/check-inventory.ts</code>	Real PostgreSQL inventory access and concurrent operation invariants	Development/disposable data only.
<code>scripts/check-demand-report.ts</code>	Report filtering, aggregation, access and privacy-safe export	Development/disposable data only.
<code>scripts/check-built-server.mjs</code>	Starts the built app and checks public/private CMS routes	Explicitly restricted to the disposable CI database.
<code>scripts/audit-live-seo.mjs</code>	Public route/sitemap/asset technical crawl	Public read-only crawl; follow its configured target and limits.
<code>scripts/database-backup.mjs</code>	Encrypted backup or isolated restore rehearsal	Separate private configuration and intended direct hosts; not an in-place restore tool.

What CI currently checks

The pinned GitHub workflow runs on push and pull request, cancels superseded runs for the same ref, and grants read-only repository contents permission. Its Linux runner uses Node 22 and a disposable PostgreSQL 17 service with synthetic CI credentials. No hosted database secrets are required.

It performs a clean install; unit/permission checks; TypeScript; all-severity dependency audit; fresh migrations into disposable Postgres; stock/report database regressions; a production build with real CMS integration enabled against that disposable database; and built-server route checks. The smoke script checks English/Arabic catalogue, staff sign-in surfaces, admin login, and denial of anonymous SKU/photo/inventory/report APIs.

The current workflow does not run every standalone integration script on every push, does not send real email, does not perform a complete device/accessibility audit and does not prove hosted database migration status. Earlier isolated integration evidence covers additional services but should not be relabelled as automatic coverage in the workflow.

Recorded final application evidence is GitHub run 36584827754 on 7f73267, with 142 unit checks, TypeScript, zero audit findings, fresh migrations, stock/report regressions, build and built-server checks. Production deployment is `dp1_4iYTN8joqJgEFHa7L2GZzR4pn2bA`; the stable site remains <https://el-amal-sigma.vercel.app>. These identifiers establish the documented checkpoint, not the result of a fresh deployment during this handbook task.

Release and rollback discipline

Vercel has a separate automatic Git integration. A push to its connected branch can change the review/production site even if the GitHub quality workflow is still running or later fails. The existing workflow is not a protected deployment gate. Before pushing a behavior change, understand that effect and arrange the intended preview/promotion process.

For an authorized release: review the diff and secret exclusions; run focused verification; verify migration compatibility and backup needs; prepare the candidate in the correct existing project; check its actual runtime; then promote/switch as intended and record the exact commit/deployment. Check public

English/Arabic pages, staff login, anonymous private boundaries, configured flags and relevant feature flows. Source commit, migration state and deployed application identity should all be recorded together.

The earlier 9a09e82 release exposed a runtime import-map URL problem after a successful build. It was immediately rolled back, repaired with an explicit path and checked as a protected candidate before promotion. This is why build success and a Vercel Ready label are insufficient. It also demonstrates an application rollback, not a database restore.

During rollback, preserve the database and review whether the older code can use the current schema and secrets. Do not automatically run down migrations. A source rollback cannot undo emails already sent, restore deleted data or reconstruct an encryption key. Verify the chosen older deployment's configuration before switching it live.

Practical troubleshooting

Symptom	First checks	Correct response
Public catalogue unavailable or empty	Intended catalogue source, CMS gate, database connectivity, published/reviewed records	Repair the real dependency or record state. Do not silently enable demo fallback.
Admin login fails	Correct environment account, five-attempt lock, active deployment, cookie/HTTPS behavior	Use the right private credential source and respect lock timing; do not repeatedly guess or bootstrap another owner. Recovery email is currently disabled.
Public submission unavailable	Configuration gates, exact origin, sender/provider and fresh durable mail health	Keep intake closed until real prerequisites work. GET capability alone is not full health.
Retry gives a request conflict	Same request UUID with changed contact/items	Return to review and create a deliberate new request identity; do not mutate saved history.
Submission saved but no staff email	Customer confirmed status, queue state, retry age and provider/scheduler readiness	Distinguish saved, queued, provider-accepted and received. Do not manually mark customer verified to clear the queue.
Confirmation link invalid	Expired hour, replaced generation, already used or malformed link	Use an eligible resend receipt or staff review; no public arbitrary-email lookup endpoint exists.
Resend control vanished	Page reload or tab closed	Receipt is intentionally memory-only. It is not recoverable from the public enquiry reference alone.
Upload area vanished after confirmation	Reload lost in-memory grant	There is no later attachment portal or token replay path. Treat replacement access as future workflow scope, not a current button.

Symptom	First checks	Correct response
Photo rejected	JPEG/PNG signature/decoder, byte/pixel limits, count, grant expiry, logical quota	Use an eligible image and unchanged retry identity. Do not rename a PDF as an image or raise limits ad hoc.
Photo download returns 404	Owner/sales session, 30-day expiry, original encryption key	Generic denial is intentional; investigate privately. Do not expose ciphertext/public URLs for troubleshooting.
Operations returns maintenance but intake stays off	No active mail transport or insufficient schedule	Expected separation: daily housekeeping never fabricates successful mail health.
Operations repeatedly overlaps	Active two-minute lease or slow/overlapping scheduler	Check timestamps and duration. Preserve leases; avoid resetting them during an in-flight send.
Staff job failed or older than 23 hours	Attempts, provider receipt and confirmation timing	Reconcile before an explicitly new delivery action; never extend its original retry window by editing dates.
Inventory refuses a hold	Exact SKU, active definition, verified enquiry/line, available/unblocked stock and optional freshness	Resolve the business facts and correct service action. Catalogue labels cannot supply the missing quantity.
Demand CSV is empty	Date range, verification/source/status filters and actual saved enquiries	Empty real demand is valid. Do not populate fictitious customer records.
Backup cannot start	Matching Postgres tools, direct connection, private key configuration, new output filename	Fix configuration privately. Never print a connection URL or key to diagnose it.
Restore rejects archive or target	Matching manifest/key, archive authentication, different allowed development host	Stop and investigate; do not bypass authentication or destination restrictions.
Deployment is Ready but routes fail	Built/runtime logs, migration state, import-map path, env scope	Verify runtime before promotion and use compatible rollback if needed.

Evidence and source index

This chapter reconciles current code with current progress. Older runbooks preserve development history and sometimes say later-completed work is pending. The latest milestone in `PROGRESS.md` and the source revision above take precedence for present status. In particular, `docs/operations-release-2026-09-29.md` retains earlier test counts/audit findings and earlier backup status; `docs/continue-project.md` still ends with a stale instruction not to claim a restore rehearsal. The actual restore report and current recovery runbook establish the later completed rehearsal.

Subject	Primary repository evidence
Current release, integration/deployment state and limitations	PROGRESS.md, README.md, docs/phase-status.md, docs/final-client-inputs.md
Runtime versions, scripts and overrides	package.json, package-lock.json
Payload collections, database adapter and disabled GraphQL	src/payload.config.ts, src/cms/collections.ts
Role and publication predicates	src/lib/access.ts, src/lib/inventory.ts, src/lib/demand-report.ts
Public data boundary and pagination	src/lib/load-catalogue.ts, src/lib/public-catalogue.ts, src/lib/read-catalogue-records.ts
App API exports	src/app/(frontend)/api/**/route.ts, src/app/(payload)/api/[...slug]/route.ts
Staff passwords/recovery/reset concurrency	src/lib/staff-security.ts, staff-recovery.ts, staff-email.ts, staff-reset-lock.ts; scripts/check-staff-and-photos.ts
Inherited staff-unlock behavior	src/cms/collections.ts; installed node_modules/payload/dist/collections/config/defaults.js, auth/defaultUnlockAccess.js, auth/operations/unlock.js
Enquiry contract/snapshot/transaction	src/lib/enquiry-preview.ts, basket.ts, enquiries.ts, submit-enquiry.ts; src/cms/enquiries.ts
Public intake gates, quotas and receipts	src/lib/customer-readiness.ts, customer-http.ts, customer-service.ts, customer-receipt.ts, customer-quota.ts, request-limits.ts, queue-capacity.ts
Confirmation and customer outbox	src/lib/enquiry-verification.ts, verification-http.ts, verification-runtime.ts, verification-message.ts, verification-outbox.ts, verification-email-worker.ts; src/cms/verification.ts, verification-emails.ts
Email transport/staff worker	src/lib/mail-transport.ts, notification-http.ts, notification-worker.ts; src/cms/notifications.ts
Scheduler, health and retention	src/lib/delivery-operations.ts, delivery-runner.ts, delivery-store.ts; src/cms/delivery-operations.ts; vercel.json
Photos and download	src/lib/enquiry-attachments.ts, attachment-http.ts, attachment-service.ts; src/cms/attachments.ts; src/components/verify-enquiry.tsx, enquiry-photo-upload.tsx
Headers and indexing boundaries	next.config.mjs, src/lib/site-policy.mjs
Migrations	src/migrations/index.ts, all registered migration files
Backup/rehearsal implementation	scripts/database-backup.mjs, scripts/lib/backup-archive.mjs, docs/database-recovery.md

Subject	Primary repository evidence
Actual backup evidence, local ignored files	artifacts/backups/el-amal-2026-09-29-verified.enc, corresponding .manifest.enc and .restore-4a77739a69f9443bbfa882dbfbda724b.json
CI and runtime smoke	.github/workflows/quality.yml, scripts/check-built-server.mjs
Secret/artifact exclusions	.gitignore, .vercelignore, .env.example (names and safe reference structure; no secret values reproduced here)
Focused operating runbooks	docs/customer-intake.md, notification-queue.md, verification-email-outbox.md, private-enquiry-photos.md, delivery-operations.md, inventory-operations.md, demand-reporting.md

Technical mechanisms have been described from source; live status is attributed to recorded evidence. No new penetration test, mail receipt test, offsite recovery test or production mutation occurred as part of writing this chapter.

Remaining work and ownership

The catalogue photographs have been supplied and all 151 main page model groups are published. They are not a remaining input. The receiving email has also been supplied and confirmed. The table below is the current work register; it separates configuration, implementation, review and explicit deferral.

Item	Current state	Needed from Nour or client	Work for developer
Customer and staff email	Code deployed but disabled	Owned sending domain or address and provider authorization	Configure sender, schedule workers, test delivery and retries, then enable intake and recovery.
Frequent worker	Daily maintenance exists	Accept required account terms for an authorized scheduler	Provision, authenticate and verify frequent executions and stale health behavior.
Exact stock	Tools built with zero exact production SKUs	Offered configurations, opening counts, hold policy, staff roles and freshness decision	Validate and import real SKUs, enter audited opening stock and rehearse staff operations.
Larger documents	JPEG and PNG only	Confirm document requirement, service authorization and retention policy	Implement private direct upload, quarantine, scanner, authorized downloads and failure tests.
Automatic offsite backup	Local encrypted backup and restore rehearsal complete	Approved destination, account access, retention and recovery objectives	Schedule copies, protect keys, verify freshness, alert failures and rehearse restoration.
External monitoring	Deferred by Nour	Revisit only when requested	Activate approved service and alerts when authorized.

Item	Current state	Needed from Nour or client	Work for developer
Company details	Provisional or hidden where facts are missing	Phone, WhatsApp, address, hours, final logo, company imagery and approved facts	Populate, link, validate bilingual presentation and structured data.
WIKI relationship	Claim component guarded until evidence exists	Genuine evidence and approved English and Arabic wording	Verify source and enable truthful relationship presentation.
Domain and search ownership	Stable Vercel URL works	Final domain ownership and Search Console access or verification token	Connect DNS, update canonical base, verify redirects and submit sitemap.
Content acceptance	Extracted and linked real catalogue	Technical, translation, image rights and business content acceptance	Correct specific findings and recheck affected records.
Staff credentials	Login works; new password policy implemented	Choose a strong unique owner password and final staff assignments	Assist secure configuration and validate recovery once mail works.
Accessibility and performance	Sampled browser and lab evidence	Client device review and acceptance	Broaden assistive technology checks, repeat representative lab tests and observe real traffic.
Interaction analytics	Private enquiry demand reporting exists	Confirm analytics and privacy choices if visitor tracking is wanted	Implement approved event measurement, separate test traffic and reconcile lead counts.
Final launch	Review site live	Approve business policies, visual presentation and operational workflow	Complete end to end checklist and record launch evidence.

No payment, purchase or acceptance of service terms should be inferred from the existence of a configuration file. Integration authorization and a live operational check must precede claims that a service is active. Monitoring remains deliberately outside the current active work.

Work that can continue without new client facts

The developer can correct verified code defects, stale operational descriptions and inconsistent runbooks; improve current interface accessibility and measured public performance; strengthen tests for existing protection boundaries; and prepare evidence and operating procedures. Those actions use the existing architecture and the approved design. They do not require fabricated company information or speculative stock.

Real email activation requires a sender and working schedule. Real stock allocation requires real configurations and counts. Search Console ownership requires control of the property. The developer can prepare these tasks but cannot replace those inputs with invented values or claim a live result based only on code.

How a stronger final website is achieved

The next substantial visual improvement comes from genuine company identity: approved logo, real people or facility photography where appropriate, truthful relationship evidence and useful technical content. Repeated decorative effects cannot replace those materials. The current motion should remain subordinate to model finding, document reading and quotation preparation.

SEO improvement depends on accurate original business content, technically correct pages, consistent entity details, useful product selection material and actual search performance review. The technical foundation is substantial; rankings depend on competition, authority, relevance, content and search systems outside this project. No top ten position or regional design award can be guaranteed.

Security work must continue throughout operation through dependency updates, access review, backup verification and incident handling. Zero known package advisories on one date is a useful result, not a permanent guarantee. Staff password quality, correct roles and protection of provider accounts remain essential.

Operator routines

At the start of a working day

1. Open the stable public site and check that the catalogue and a representative product load.
2. Sign in through /admin with your own staff account. Use the role assigned to your job.
3. Sales reviews new or changed enquiries once real intake is activated. An empty report before activation is expected.
4. Warehouse staff review held, blocked and expired stock through /staff/inventory. Compare physical counts when policy requires it.
5. The owner reviews delivery operations once mail is active. Successful daily maintenance alone does not prove customer email delivery.

When publishing a catalogue change

1. Keep the manufacturer's model identity and documentary source together.
2. Review both languages, application classification, image path, specification rows and downloads before publication.
3. Save the permitted CMS fields and resolve publication validation errors rather than bypassing them.
4. Open the public English and Arabic product pages using the site's own links. Confirm the image, table, availability date and technical document destination.
5. Check a model search and the relevant category. If the change affects canonical URLs or categories, run the SEO checks for those routes.

When recording stock

1. Identify the exact existing SKU. Do not use a generic product family where range, connection or material differs.
2. Choose the correct operation and enter a meaningful reference or reason. Receipts add units; dispatch consumes a reservation and removes physical units; release only returns held units to availability.
3. Check the result before repeating a request. The service prevents the same request from moving stock twice, but a newly created request is a new instruction.
4. Investigate physical discrepancies and record a justified adjustment. A count confirmation is not a substitute for an adjustment.
5. Preserve the audit history. Posted ledger entries are not edited to hide a mistake; corrections are additional explained events.

At each release

The developer reviews the diff, migration implications, tests and package advisories, then verifies the actual deployed pages and private boundaries. A build marked Ready is not sufficient. The project previously detected a build that succeeded but failed at runtime, rolled it back and added actual server startup checks. Keep a known working deployment available for recovery.

Periodically

Review who has access, test restore procedures in an isolated database, inspect backup age, review error and mail health, check broken manufacturer links, recheck measured performance and assess new dependency advisories. The exact business schedule and responsibility should be agreed with the owner; an automated timetable is not claimed where no service has been configured.

Troubleshooting by symptom

Symptom	Likely meaning	Safe next action
Admin says credentials are incorrect	Wrong identity or password, logout, or account state	Confirm the staff email, wait for any lockout window, then contact the owner. Do not share the password in chat. Recovery email is unavailable until activated.
Staff page redirects to login	Session expired or missing	Sign in again at /admin and return to the staff URL.
Staff page or API refuses access	Current role does not permit that operation	Ask the owner to review role assignment; do not bypass the endpoint.
Product cannot be published	Required bilingual content or review gate is incomplete	Review the CMS validation message and the publication checklist.

Symptom	Likely meaning	Safe next action
Image fails after editing details	Image path does not match a deployed public asset	Use a verified repository asset path and ask the developer to add new files. There is no general Media upload collection.
Customer cannot submit an RFQ	Public intake is deliberately disabled or readiness has failed	Check sender, worker health and intake configuration. Do not tell the customer a request was received if it was not saved.
No email arrives	Sender or worker is not active, or queue/provider failure	Owner inspects the correct queue and delivery operations. Do not reset immutable enquiry snapshots.
Photo upload is unavailable	Verification is incomplete, capability expired, flag is off or limits were exceeded	Complete the supported verified journey after mail activation; check file type, size and count.
Cannot reserve stock	No exact SKU, insufficient available units, stale count, unauthorized role or ineligible enquiry	Read the refusal and correct the underlying record or policy. Model labels do not prove counted availability.
Demand report is empty	No matching persisted enquiries for selected dates and cohort	Check Cairo date boundaries and filters. It is not a website traffic report.
Google does not show a page	Technical indexability does not guarantee indexing	Verify the canonical URL and Search Console once property access is configured.
Client sees a regression after deployment	Code, configuration, migration or runtime problem	Capture URL and time without secrets; verify runtime logs and use the known working deployment while investigating.

Source ownership and continuation

The project is saved in C:\Users\noura\OneDrive\Documents\ChatGPT\EL-AMAL 4 and on the codex/el-amal-foundation branch of the EL-AMAL GitHub repository. Source, migrations, tests and this handbook are versioned. node_modules and build output can be regenerated. Live database contents, credentials, ignored catalogue originals and test artifacts are separate assets.

A new assistant or developer should read AGENTS.md, PROGRESS.md, docs/phase-status.md, this handbook and the current Git diff before acting. Do not restart the catalogue extraction. Do not ask again for the receiving inbox or the supplied 29 pages. Keep production and development database connections separate and never run development mutation checks against production.

Signing in to another ChatGPT account does not itself move or back up the database. Access from another device depends on an already configured host connection or a checked out repository with securely supplied environment settings. An unrelated cloud chat does not automatically have this laptop's local files. No automatic resume task is scheduled by this handbook.

Glossary

Term	Plain meaning
CMS	The private content management system where staff edit permitted records.
Model or family	A manufacturer's product identity, which may include several possible configurations.
SKU	One exact supplied configuration that can be counted in inventory.
RFQ	Request for quotation, describing what a customer wants before price and supply are agreed.
Ledger	The append only history of stock movements and reviews.
Reservation or hold	Units temporarily allocated to an eligible request without leaving the warehouse.
Blocked stock	Physical units excluded from availability, for example while a staff review is needed.
Outbox	A database queue that preserves a message until a worker can send or conclusively fail it.
Idempotency	Repeating the same operation identifier and input does not apply the action twice.
Transaction	Several database changes succeed together or are rolled back together.
Migration	A versioned change to database structure that accompanies application changes.
Canonical URL	The preferred address declared for a page.
Hreflang	Links identifying the language alternatives of equivalent pages.
Structured data	Machine readable facts in a page, using Schema.org vocabulary.
Noindex	An instruction asking compliant search engines not to index a page; it is not authentication.
LCP	How long the largest visible content element takes to appear in the measured load.
CLS	The amount of unexpected visual movement during the measured page session.
INP	How responsive real interactions are over a page visit; it requires appropriate field evidence.
CI	Automated checks run when source changes are pushed.
Encryption key	A secret required to decrypt protected data; it must be stored separately from the backup.

Evidence and verification index

The completed release has 142 passing unit checks, a successful TypeScript check, a clean package advisory audit at the recorded date, real development database regressions, and a successful cloud build with actual server route checks. GitHub run 36584827754 verified application commit 7f7326704305012e9637702b3112b5c85f554833. The live deployment identifier is dpl_4iYTN8joqJgEfHa7L2GZzR4pn2bA. The stable client address remains unchanged.

The final technical crawl covered 342 public pages, 342 sitemap URLs and 155 assets with zero detected issues in the implemented checks. Representative mobile Lighthouse samples measured homepage performance from 89 to 91 and product performance 90. The final homepage sample had LCP 2.8 seconds, total blocking time 240 ms and CLS zero. The preceding product sample had LCP 2.9 seconds. These do not establish the LCP below 2.5 seconds target or real visitor INP.

The encrypted restore rehearsal recovered 24 tables and 572 rows with matching row fingerprints and constraint count into a separate temporary development database. It was performed on the backup taken before the additive inventory completion migration. The temporary database was removed. Future recovery must apply the appropriate forward migrations before switching a replacement environment into service.

The main evidence files are in the ignored artifacts/2026-09-29/completion folder: seo-final.json, live-release.json, lighthouse-final-home.json, lighthouse-home.json, lighthouse-product.json and unit-tests.log. The earlier launch checks are in artifacts/2026-09-29/launch. The published handbook summarizes those findings; screenshots and raw diagnostic files are not customer data or a substitute for operational acceptance.

The source references in the preceding chapters are repository relative. Open them within the local project or at the matching Git commit. Configuration variable names are provided for developers; values and credentials are intentionally kept in private environment settings. This handbook contains no login password, database URL, API key or backup key.

Technical references

The project code and recorded tests establish what EL AMAL actually does. These official references explain several of the underlying approaches; they do not certify this implementation.

- [Google guidance on canonical URLs](#) explains why consistent preferred addresses, internal links and sitemap entries help consolidate duplicate pages. A canonical declaration is a signal, not a guarantee that Google will select it.
- [Google guidance on language alternatives](#) explains the use of equivalent English and Arabic language links.
- [Core Web Vitals thresholds](#) explains the good experience thresholds of LCP at most 2.5 seconds, INP at most 200 milliseconds and CLS at most 0.1, evaluated at the 75th percentile of visits. A Lighthouse sample is not that field dataset.
- [OWASP authentication guidance](#) provides the security context for strong credentials, throttling, neutral login and recovery responses, and session controls. Independent penetration testing and broader authentication hardening remain separate from implementing those controls.

Update after the handbook baseline

Date: 29 September 2026. The main handbook records application release 7f73267 and was delivered before further development resumed. This dated update records what changed afterward; it takes precedence over the two source findings described in the baseline security chapters.

- **Staff account unlocking:** only an owner may unlock a staff account. Sales, warehouse, catalogue editors and anonymous callers are refused. This keeps account administration within the owner role. Existing login and password rules are unchanged.
- **Notification privacy:** normal owner and sales API reads retain useful delivery status, attempts and reference fields but omit the internal delivery key and worker lease identifier. Merely hiding a field in the admin interface did not restrict its API representation. Trusted internal workers retain access needed to send and finish a queued message.
- **Clearer admin instructions:** the enquiry screen now explains which fields staff can edit, where stock holds are managed and where email delivery results are found. The legacy Delivery Status field is explicitly described as a submission snapshot, not current email status.
- **Repeatable verification:** the new isolated database test covers eleven account, queue and worker checks. Before correction it reproduced five failures; afterward all eleven succeeded. It also verifies that a legitimate owner can unlock and that two competing fake workers deliver exactly once. All 142 unit tests and TypeScript succeeded. The database regression is included in GitHub's existing quality workflow.
- **Continuation notes:** obsolete wording saying that a restore rehearsal had not happened was corrected. The completed rehearsal is documented separately from automatic offsite backup scheduling, which is still outstanding.

No customer data, actual stock, owner password, public design or schema was changed for these fixes. All test data used an empty development schema that was removed afterward. Email tests used a fake transport. Customer sending and recovery delivery remain disabled pending their existing prerequisites; monitoring remains deferred at Nour's request.

Release verified: application commit b3408a500c38dcc39e70a13a7ee68393ba50365d is live at the stable client URL as deployment dpl_8YSXSLAXwnrE3B8B1xEyHXs6J2JD. [GitHub quality run 36606118269](#) succeeded, including the new database regression, production build and server startup checks. Current npm audit reports zero known advisories.

Live checks verified English/Arabic home and product pages, anonymous private-route and unlock rejection, owner permissions, hidden notification fields, updated enquiry instructions and the staff reports screen. Checked 390px screens had no horizontal overflow or browser JavaScript errors. The initial API test omitted the Origin header required by Payload's cookie authentication; the corrected request passed without weakening that protection. No production account unlocking or stock mutation was used as a test. This is scoped release evidence, not a full security certification or final client acceptance.